JOINING PROCESSES

- Joining includes welding, brazing, soldering, adhesive bonding of materials.
- They produce permanent joint between the parts to be assembled.
- They cannot be separated easily by application of forces.
- They are mainly used to assemble many parts to make a system.
- Welding is a metal joining process in which two or more parts are joined or coalesced at their contacting surfaces by suitable application of heat or/and pressure.
- Some times, welding is done just by applying heat alone, with no pressure applied
- In some cases, both heat and pressure are applied; and in other cases only pressure is applied, without any external heat.
- In some welding processes a filler material is added to facilitate coalescence(Joining)

Joining Processes: Welding, Brazing, Soldering

1. Brazing and Soldering: Melting of filler rod only
   - Brazing: higher temperature, ~brass filler, strong
   - Soldering: lower temp, ~tin-lead filler, weak
2. Welding: Melting of filler rod and base metals

Advantages of welding:

- Welding provides a permanent joint.
- Welded joint can be stronger than the parent materials if a proper filler metal is used that has strength properties better than that of parent base material and if defect less welding is done.
- It is the economical way to join components in terms of material usage and fabrication costs. Other methods of assembly require, for example, drilling of holes and usage of rivets or bolts which will produce a heavier structure.

Disadvantages of welding:

- Labour costs are more since manual welding is done mostly.
- Dangerous to use because of presence of high heat and pressure.
- Disassembly is not possible as welding produces strong joints.
- Some of the welding defects cannot be identified which will reduce the strength.

Classification of Welding processes

- Arc Welding
  1)Gas tungsten arc welding(TIG) or (GTAW)
  2)Gas metal arc welding(MIG) or (GMAW)
  3)Shielded metal arc welding(SMAW)
  4)Submerged arc welding
  5)Plasma arc welding
  6)Flux cored arc welding(FCAW)

- Resistance welding
  1)Spot welding
  2)seam welding
  3)Projection welding
  4)Resistance butt welding

- Gas welding

  1)Oxy-acetylene welding

  2) Oxy-hydrogen welding

  3)Air -acetylene welding

  4)Pressure Gas welding

- Thermo chemical welding Process

  1)Thermit welding

  2)Atomic hydrogen welding

- Radient energy welding Process

  1)Electron beam welding

  2)Laser beam welding.

Types of welding:

Welding processes can be broadly classified into

(i) fusion(non-pressure) welding, and (ii) solid state welding(pressure welding)

Fusion welding:

In fusion-welding processes, heat is applied to melt the base metals. In many fusion welding processes, a filler metal is added to the molten pool during welding to facilitate the process and provide strength to the welded joint.

When no filler metal is used, that fusion welding operation is referred to as Autogenous weld.

Types: Arc welding, Resistance welding, Gas welding, electron beam welding, laser welding

Solid State Welding:

- In this method, joining is done by application of pressure only or a combination of heat and pressure.
- Even if heat is used, the temperature in the process is less than the melting point of the metals being welded (unlike in fusion welding).
- No filler metal is utilized.

Diffusion welding: Two part surfaces are held together under pressure at elevated temperature and the parts join by solid state diffusion.

Friction welding/Stir welding: Joining occurs by the heat of friction and plastic deformation between two surfaces.

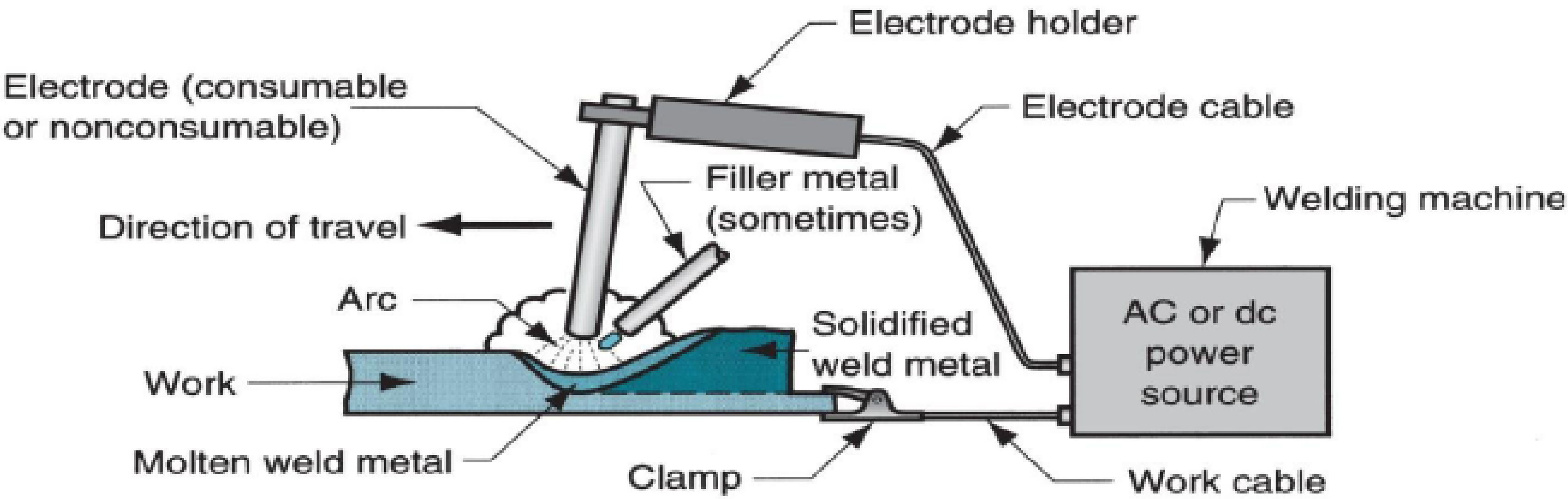
Ultrasonic welding: Moderate pressure is applied between the two parts and an oscillating motion at ultrasonic frequencies is used in a direction parallel to the contacting surfaces

Arc welding Process

It is a fusion welding process in which the melting and joining of metals is done by the heat energy generated by the arc between the work and electrode.

An electric arc is generated when the electrode contacts the work and then quickly separated to maintain the gap. A temperature of 5500°C is generated by this arc.

This temperature is sufficient to melt most of the metals. The molten metal, consisting of base metal and filler, solidifies in the weld region. In order to have seam weld, the power source moves along the weld line.



Electrodes

- Two types of electrodes are used: **consumable and non-consumable**
- **Consumable electrodes:**

Present in rod or wire form with 200 to 450 mm length and less than 10 mm diameter. This is the source of filler rod in arc welding.

The electrode is consumed by the arc during the welding process and added to the weld joint as filler metal.

The consumable electrodes will be changed periodically as it is consumed for each welding trials. This becomes a disadvantage for welder and reduces the production rate.

- **Non-Consumable electrodes:**

The electrodes are not consumed during arc welding. Though this is the case, some depletion occurs because of vaporization.

Filler metal must be supplied by means of a separate wire that is fed into the weld pool.

Arc shielding

Shielding gas:

This covers the arc, electrode tip and weld pool from external atmosphere. The metals being joined are chemically reactive to oxygen, nitrogen, and hydrogen in the atmosphere.

So the shielding is done with a blanket of **gas or flux**, or both, which inhibit exposure of the weld metal to air.

Common shielding gas: **Argon, Helium**

Flux:

Used mainly to protect the weld region from formation of oxides and other unwanted contaminants, or to dissolve them and facilitate removal.

During welding, the flux melts and covers the weld region giving protection and it should be removed by brushing as it is hardened.

Additional function, other than giving protection: stabilize the arc, and reduce spattering

Power source in arc welding:

Both AC and DC can be used; DC is advantageous as better arc control is possible.

Polarity:

Straight polarity in which work piece is positive and electrode is negative is suitable for shallow penetration (like in sheets) and joints with wide gaps.

Reverse polarity in which work piece is negative and electrode is positive is suitable for deeper welds.

Arc welding processes with consumable electrodes

• **Shielded metal arc welding (SMAW):**

- In this process, a consumable electrode consisting of a filler metal rod which is coated with chemicals that provide flux and shielding, is used.
- Generally the filler metal has chemical composition very close to base metal.

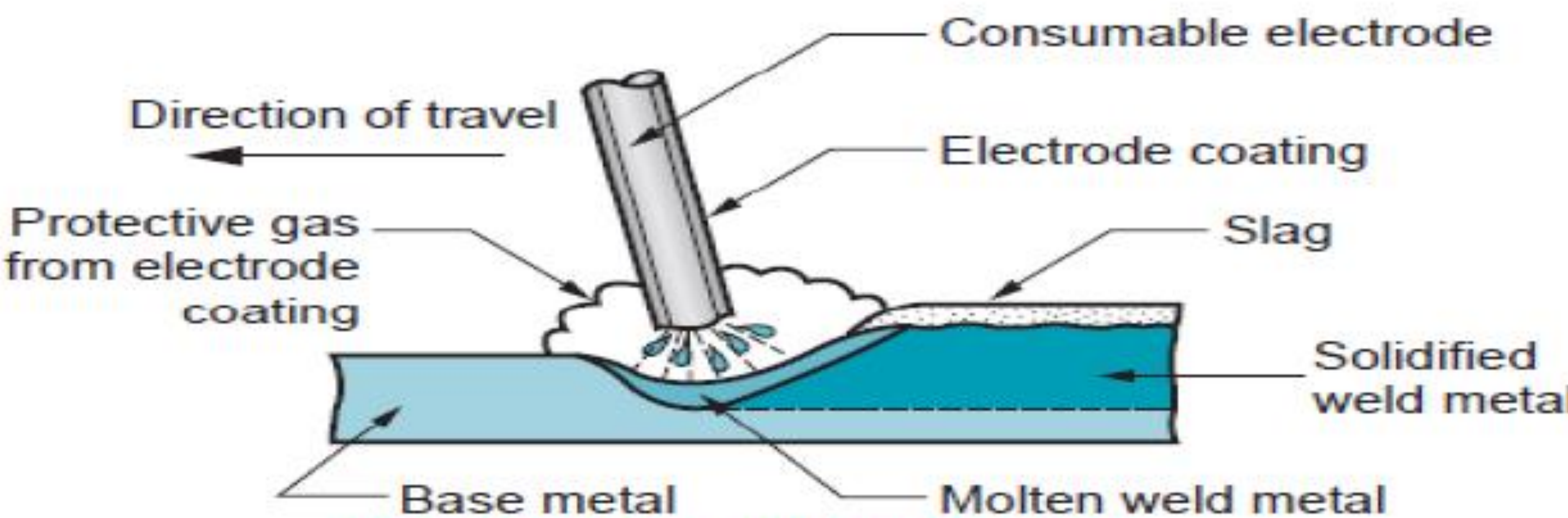
- **Filler rod coating**: Coating consists of powdered cellulose (cotton and wood powders) mixed with oxides, carbonates, combined using a silicate binder.
- This coating provides protective layer to the weld pool and stabilizes the arc.

- current: < 300 A; Voltage: 15 – 45 V.

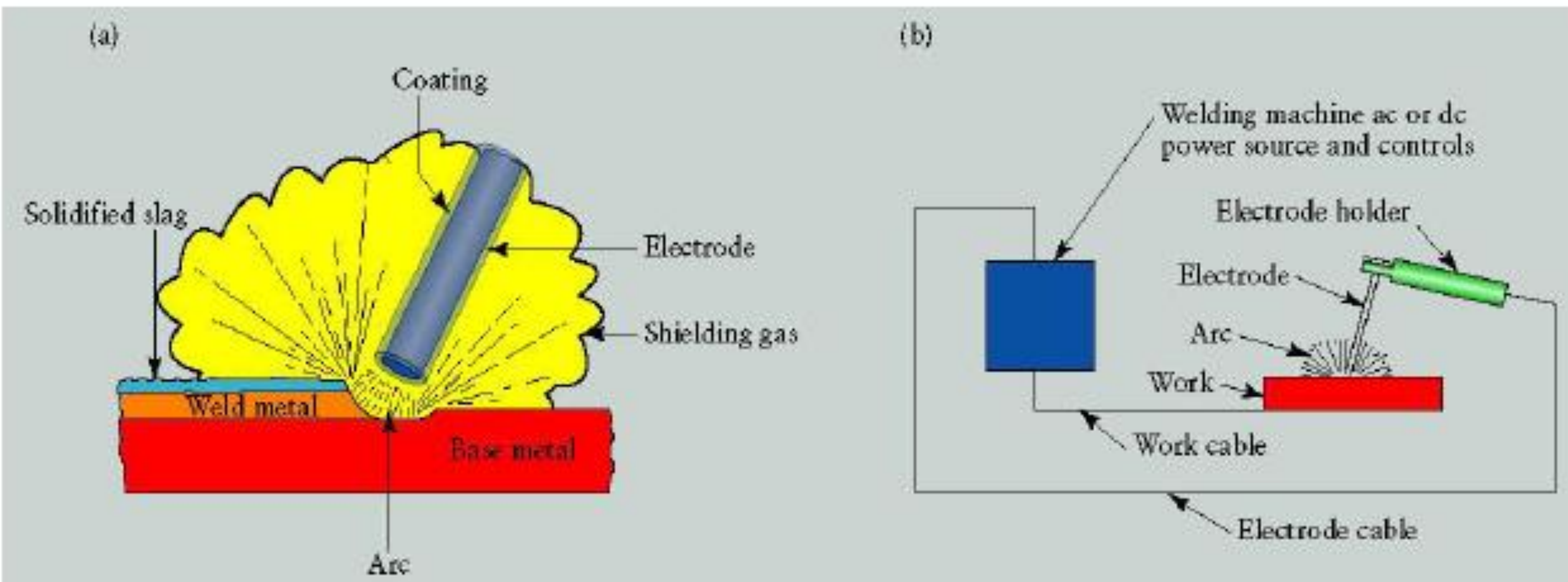
- **Applications:** ship building, construction, machine structures etc.

- **Materials:** grades of steel, stainless steel etc. are welded. Al, Cu, Ti alloys are not welding using SMAW.

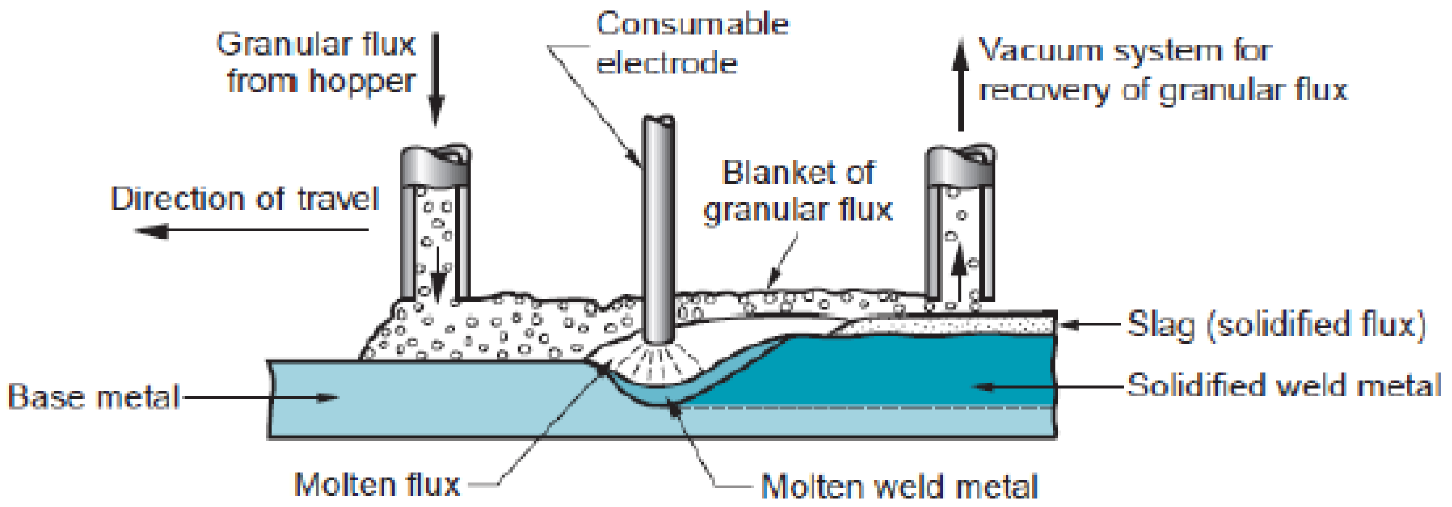
- **Disadvantages**: repeated change of electrodes, current maintained in typical range



Shielded metal arc welding (SMAW)



Submerged arc welding (SAW):



- In submerged arc welding also known as **hidden arc welding, submerged melt welding** or **sub-arc welding** the arc is struck between a metal electrode and the work piece under a blanket of granular flux.

-The welding action takes place under the flux layer without any visible arc, spatter, smoke or flash

-In this process, a continuous bare electrode wire is used.

-The shielding is provided by external granular flux through hopper.

- Granular flux is provided just before the weld arc.

- granular flux completely provides protection from sparks, spatter, and radiation and hence safety glasses, gloves can be avoided.

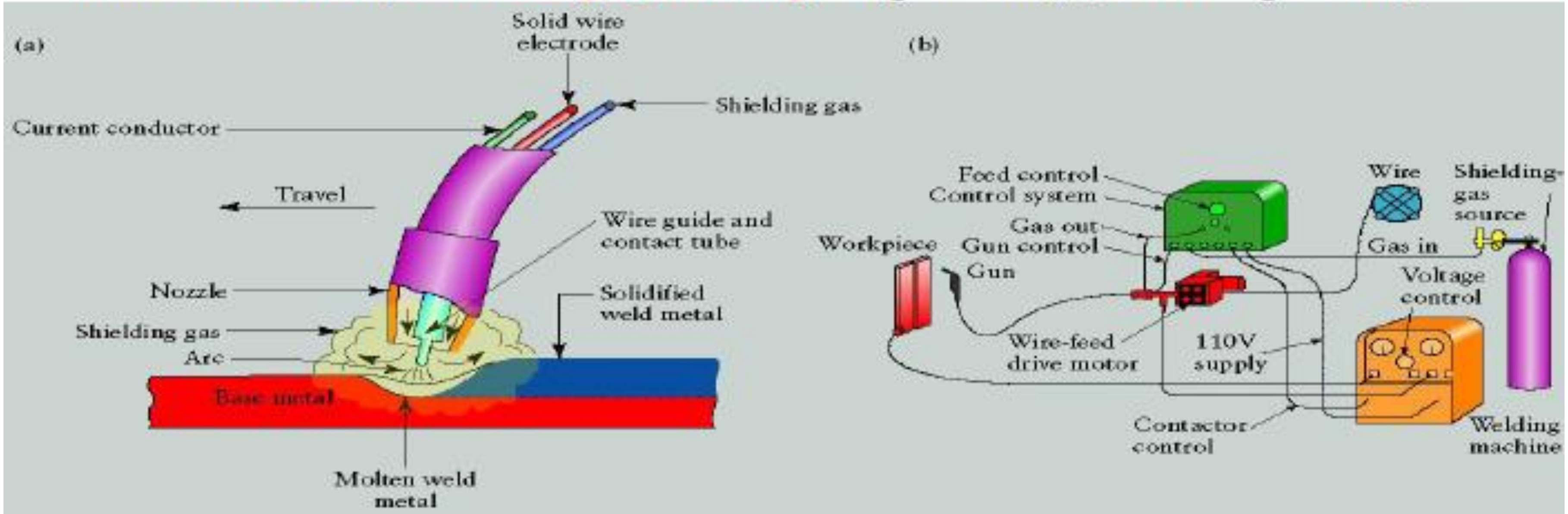
-some part of flux gets melted and forms a glassy layer.

Application:

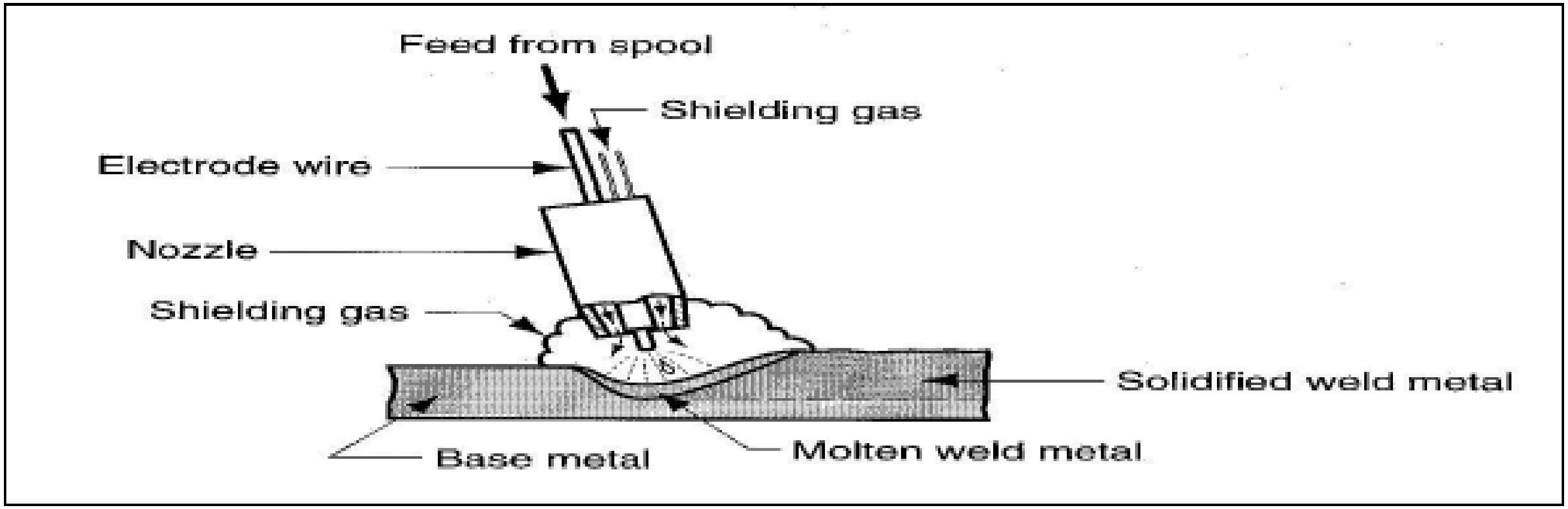
longitudinal and circumferential welds for large diameter pipes, tanks, and pressure vessels; welded components for heavy machinery. Steel plates of 25 mm thick are welded

Gas metal arc welding (GMAW) OR (MIG)

- In this process, electrode is a consumable wire (0.8 to 6.5 mm diameter).
- shielding gas is provided separately over arc by a pipe
- Shielding gas: Helium, Argon, mixture of gases; used mainly for Al alloys.
- active gases like CO_2 is used for welding steel grade material.
- As compared to SMAW, GMAW can be used for multiple weld passes as there is no deposition of slag and hence no brushing involved. (advantage)
- advantage: automation of welding possible as continuous weld wires are used, and not sticks as in SMAW.
- Also called MIG (metal inert gas) welding, CO_2 welding (when CO_2 is used).



- **Gas metal arc welding (GMAW),** sometimes referred as metal inert gas **MIG welding**, is a welding process in which an electric arc is formed between a consumable wire electrode and the work piece metal(s), which heats the work piece metal(s), causing them to melt, and join.
- Along with the wire electrode, a shielding gas is fed through the welding gun, which shields the process from contaminants in the air.
- A constant voltage, direct current power source is most commonly used with GMAW.
- shielding gases are necessary for gas metal arc welding to protect the welding area from atmospheric gases such as nitrogen and oxygen, which can cause fusion defects, porosity.



Arc welding processes with non consumable electrodes

Gas Tungsten Arc Welding (GTAW):

- It uses a non-consumable tungsten electrode and shielding gas (inert gas) for shielding.
- Also called tungsten inert gas welding (TIG)

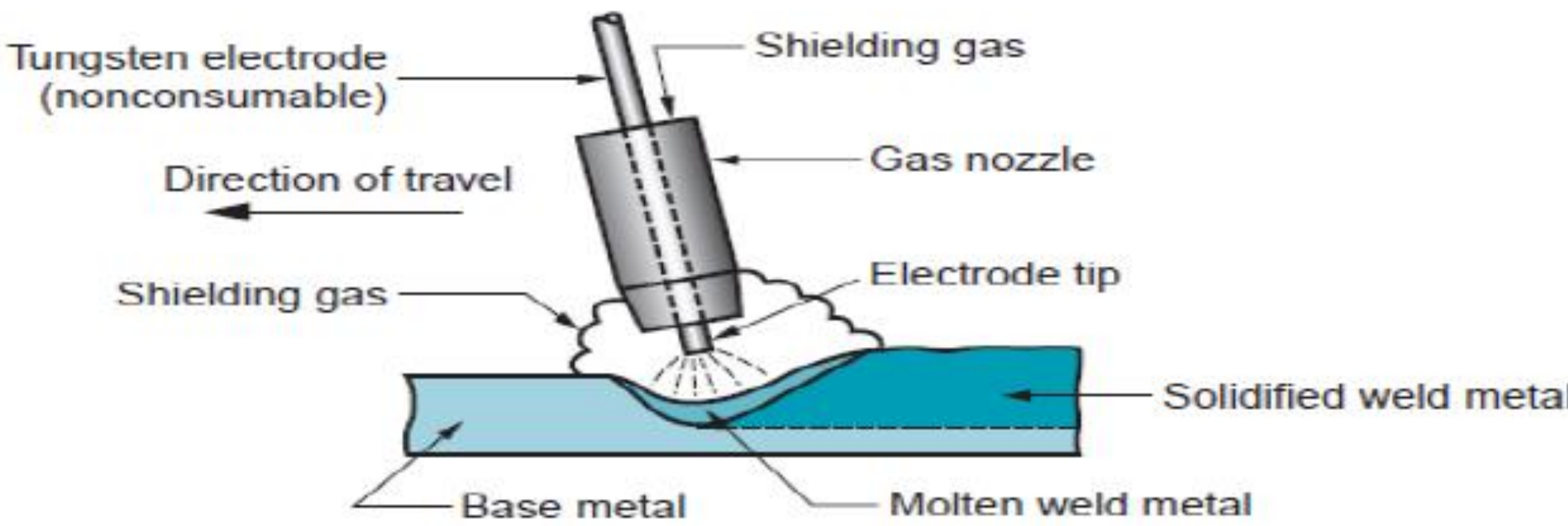
- usage of filler wire is optional and is heated by arc and not transferred across the arc.

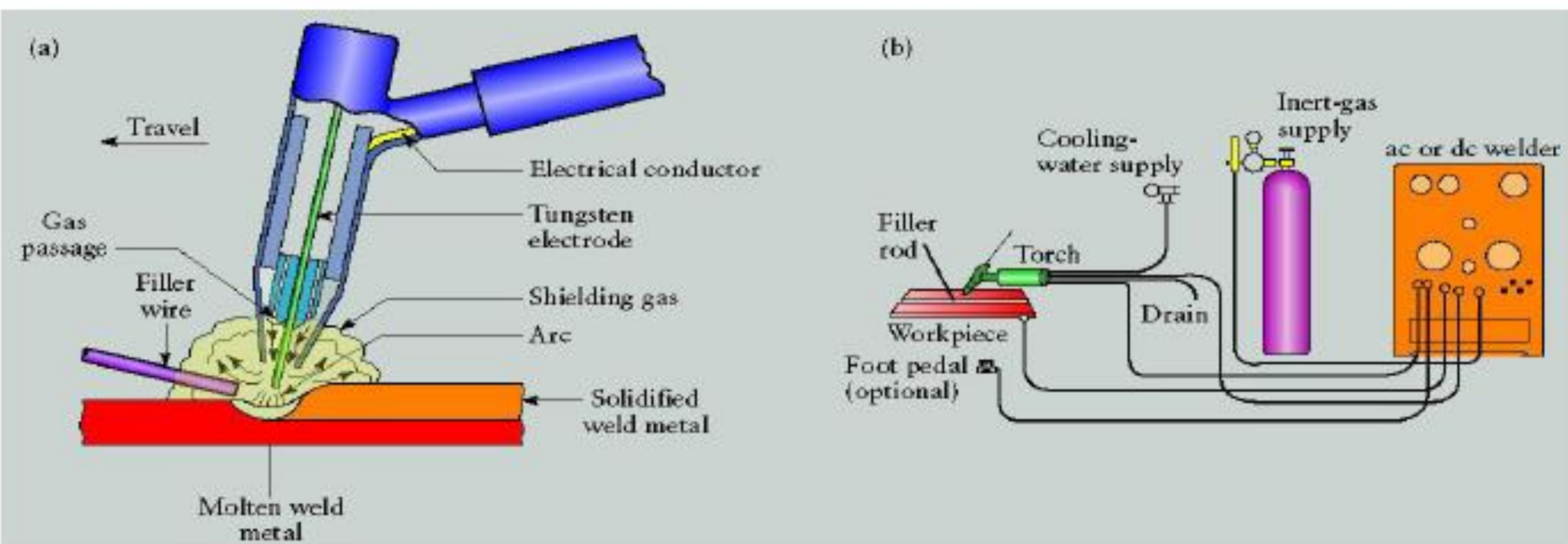
- Tungsten is a good electrode material due to its high melting point of 3400°C.

- Advantages: high quality welds, no weld spatter because no filler metal is transferred across the arc.

- High skill level required to achieve good weld

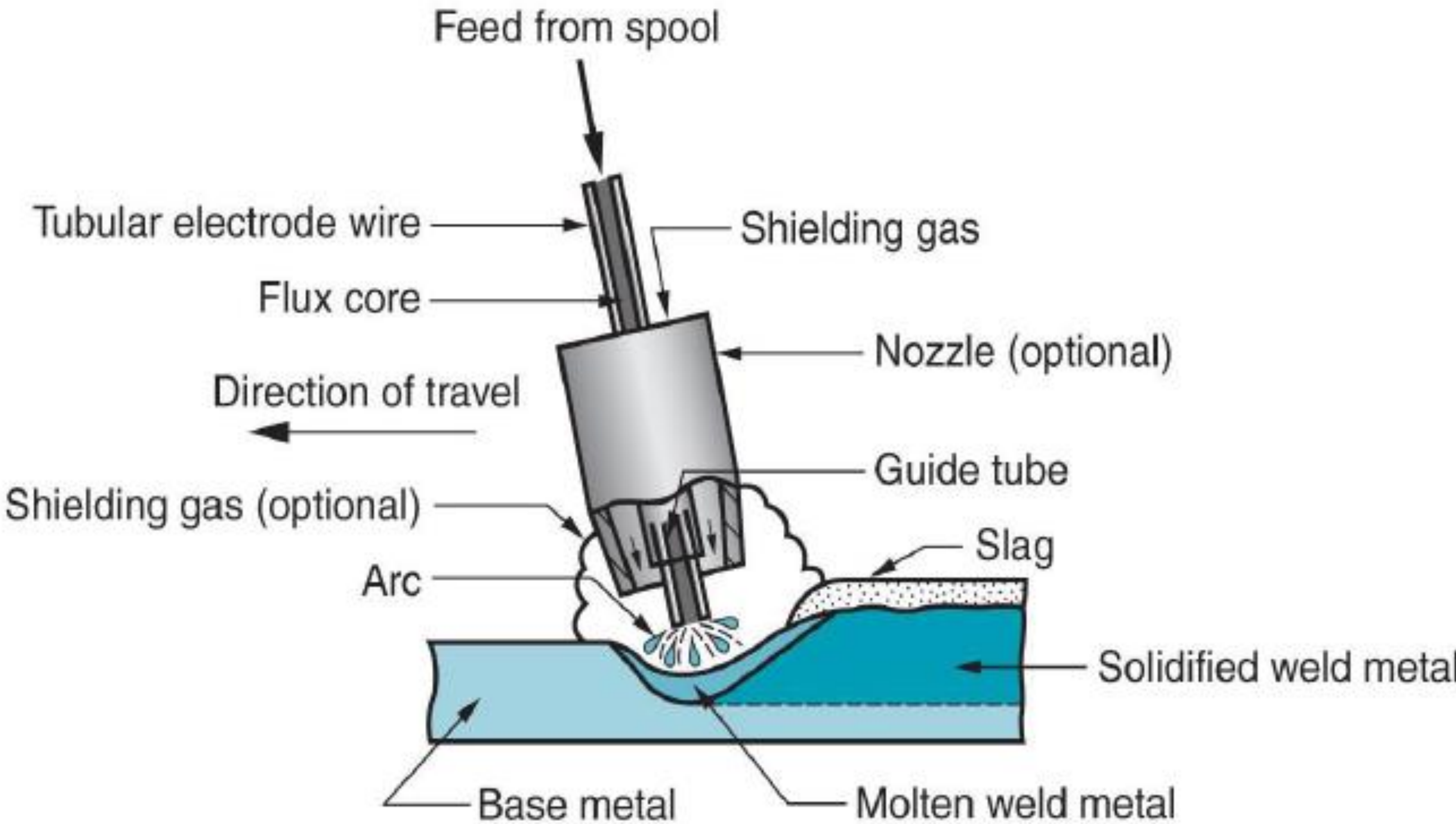
- Difficult to automate


Tungsten electrode
(nonconsumable)
Shielding gas
Gas nozzle
Direction of travel
Electrode tip
Shielding gas
Solidified weld metal
Base metal
Molten weld metal


(a)
Travel
Electrical conductor
Tungsten
electrode
Gas
passage
Filler
wire
Shielding gas
Arc
Solidified
weld metal
Molten weld
metal
(b)
Inert-gas
supply
Cooling-
water supply
ac or dc welder
Filler
rod
Torch
Drain
Workpiece
Foot pedal
(optional)

Flux Cored Arc Welding (FCAW)

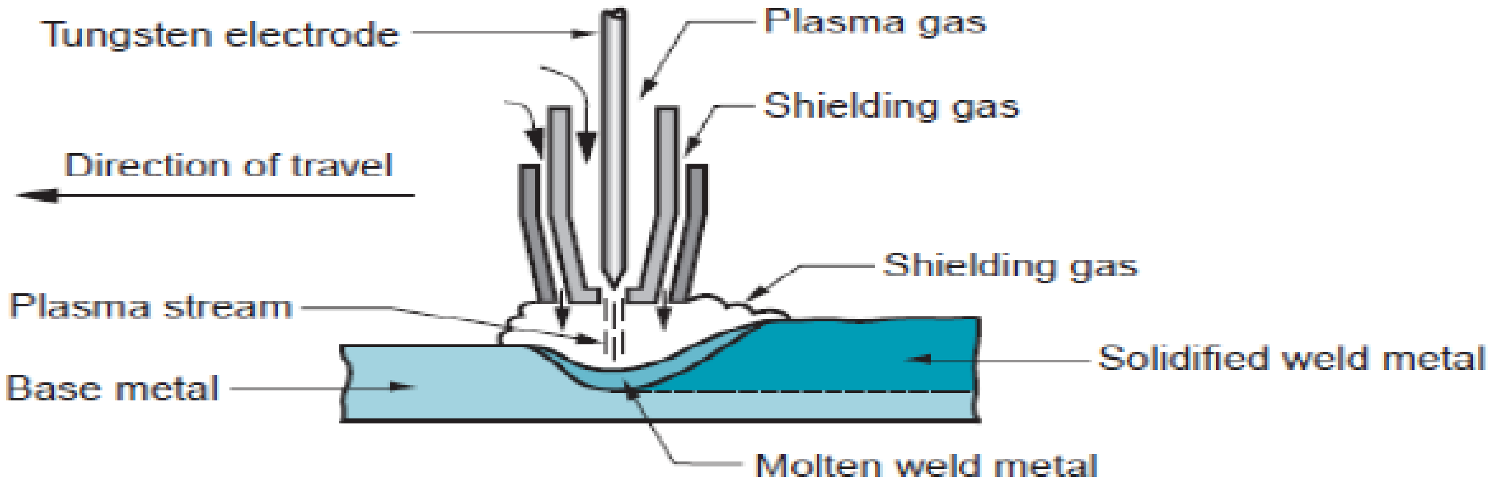
- FCAW is an arc welding process in which the heat for melting is obtained from an arc between a hollow consumable tubular electrode and the work piece.
- **The hollow electrode is filled with flux. That is why the process is called flux cored arc welding process.**
- This process is economical compared to SMAW and GAW.
- Metals over a wide range of thickness can be welded.
- Because of its higher productivity FCAW is finding wide applications for shop fabrication, field erection work and maintenance.
- It can be used for welding of cast iron, low carbon steel, low and high alloy steels and stainless steels.
- Well known applications include machine frames, bulldozer blades, bridge grinders, furnace tubes and diesel engine bodies.



Flux Cored Arc Welding (FCAW)

Plasma arc welding

- It is a variety of gas tungsten arc welding in which a constricted (narrow) plasma arc is used for welding.
- In PAW, a tungsten electrode is kept in a nozzle that focuses a high velocity stream of inert gas into the region of the arc to form a high velocity, intensely hot plasma arc stream.
- Temperatures in plasma arc welding reach 17,000°C. This is mainly due to the constriction of the arc. The input power is highly concentrated to produce a plasma jet of small diameter and very high power density.
- The process can be used to weld almost any material, including tungsten.



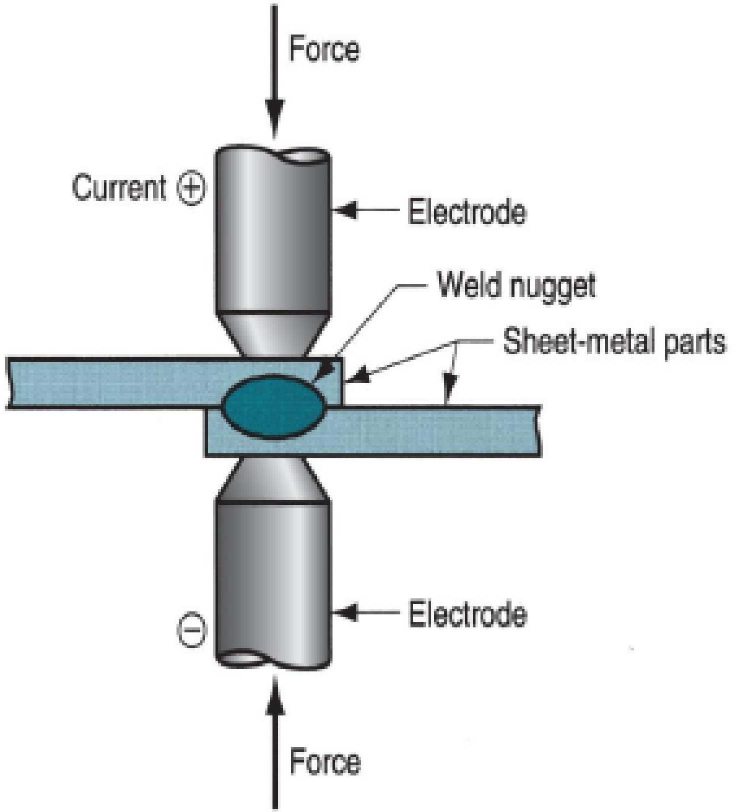
Arc welding Electrode

The Electrode and Coating

Coating is a combination of chemicals

- **Cellulosic electrodes** contain cellulose
- **Rutile electrodes** titanium oxide (rutile)
- **Basic electrodes** contain calcium carbonate (limestone) and calcium fluoride (fluorspar)

Resistance Welding



- Resistance welding is the process in which two or more parts are welded by the coordinated and regulated use of heat and pressure.
- The heat in the resistance welding process is generated by the resistance offered by the work pieces to the flow of low voltage high density electric current.
- A typical resistance welding cycle consists of (1) **squeeze** (2) **weld** and (3) **hold** period
- The resistance welding processes include spot welding, seam welding, roll spot welding, projection welding, upset welding, flash welding and percussion welding.

- Practically all metals can be resistance welded and the operation is quite fast.
- Weld nugget is generated by this process
- RW uses no shielding gases, flux, or filler metal.
- electrodes that conduct electrical power to the process are non-consumable

Current : 5000 to 20,000 A

Voltage : < 10V

Duration of current : 0.1 to 0.4 s (in spot-welding operation)

- Resistance in the welding circuit is the sum of (1) resistance of the electrodes, (2) resistances of the sheet parts, (3) contact resistances between electrodes and sheets.

- Resistance at the welding surfaces depends on surface finish, cleanliness, contact area, force. No paint, oil, dirt, and other contaminants should be present to separate the contacting surfaces.

- Advantages: no filler rod required, high production rates, automation and mechanization are possible.
- disadvantages: restricted to lap joint, costly equipment

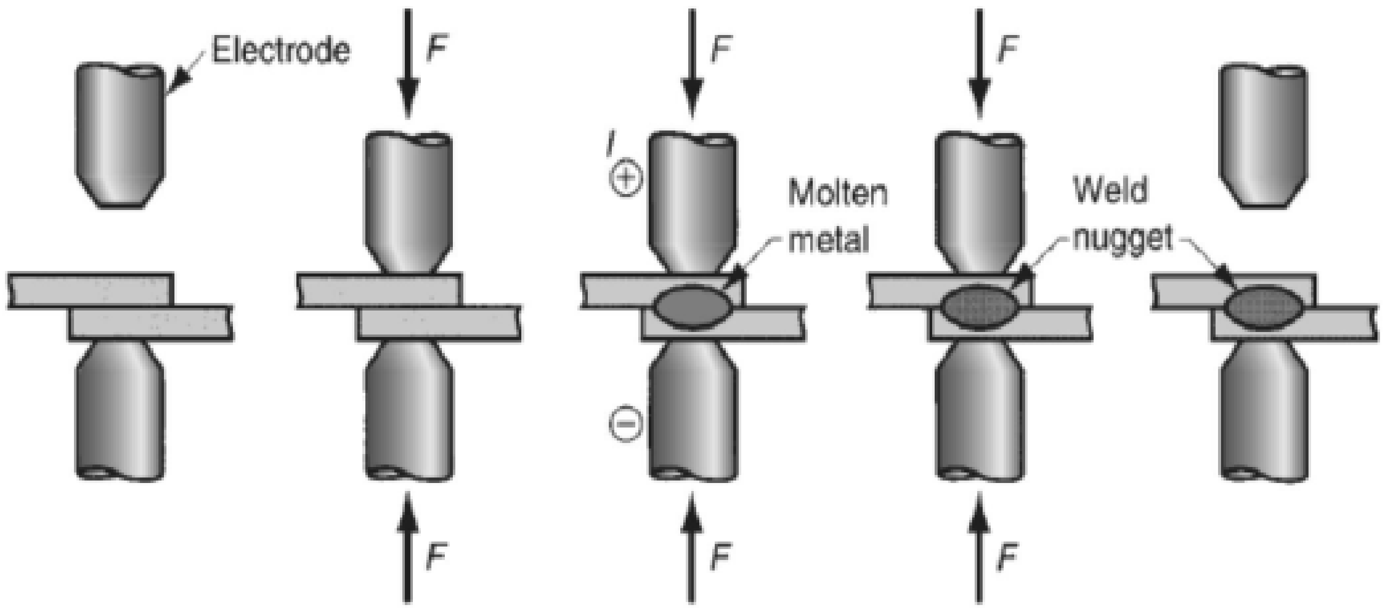
Resistance spot welding

Importance: a typical car body has got app. 15, 000 spot welds.

In this process, the fusion of electrodes is done by electrodes having opposing charges at one location. The sheet thickness has to be less than 3 mm for a good spot weld.

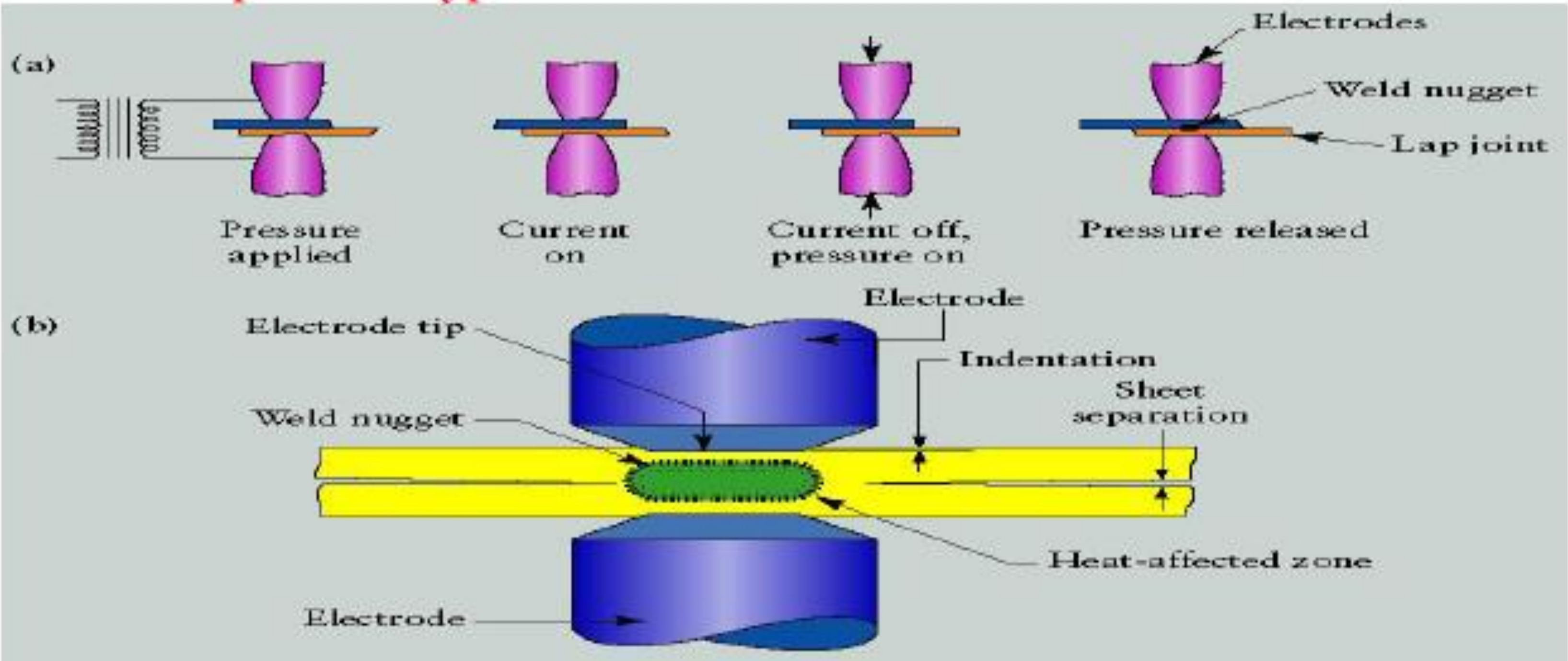
The shape of electrode tip is important like round, hexagonal, square etc. The nugget shape will be app. 5-10 mm in this case.

Electrodes in RSW: (i) Copper based, (ii) Refractory metal (Cu, Tungsten combinations)



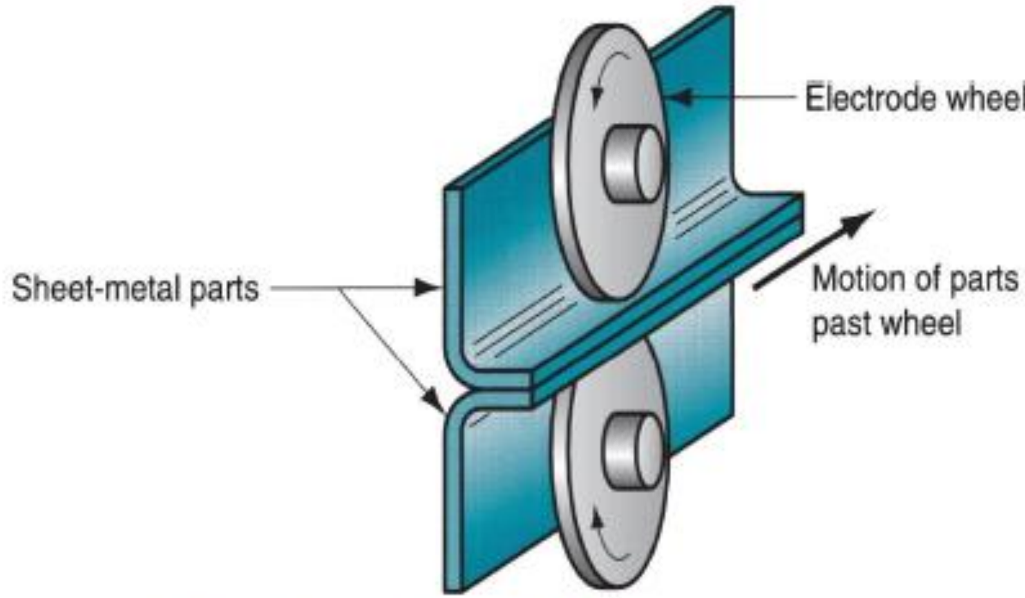
Steps in spot welding

- This is the simplest form of resistance welding and does not possesses any problem for welding sheets ranging up to 12.5 mm in thickness.
- For best results the surfaces to be welded must be free from scales and foreign matter.
- Spot welding machines are available in three different types : **stationary single-spot, portable single spot** and **multiple spot welding** machines.
- Stationary single spot machines can be further classified into **rocker arm type** and **direct pressure type**.



Seam Welding :

- In principle, seam welding is similar to spot welding except that it uses disc shaped electrodes as shown in fig.



- Unlike spot welding the disc shaped electrodes are not separated after each weld but maintain a continuous pressure over the work pieces.
- Two common types of seam weld are **Lap seam welding** and **Butt seam welding.**
- The electrode current is timed to flow in pulses so that a row of welds is produced along the interface.

Gas welding / Oxyfuel gas welding

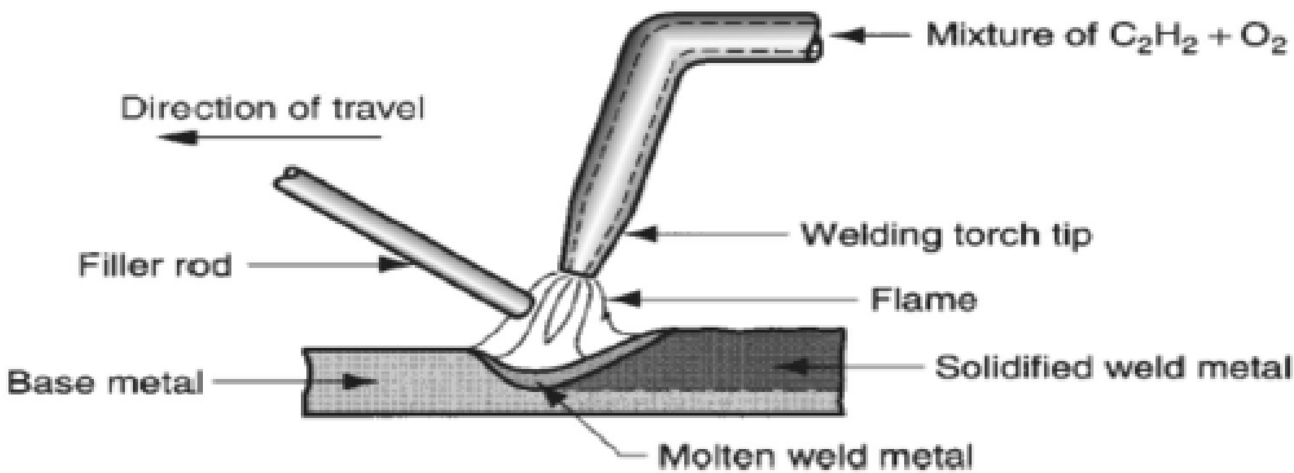
-In this process, various fuels are mixed with oxygen and burnt to perform welding. Eg: Oxyacetylene welding

➢Oxyacetylene welding (OAW):

In this case, welding is performed by a flame formed by the combustion of oxygen and acetylene. The flame comes from a torch.

A filler rod coated with flux is used sometimes which prevents oxidation, creating a better joint.

Acetylene is a famous fuel because it is capable of generating a temperature of 3500°C.



The chemical reaction between oxygen and acetylene happens at two stages as given below.

$$C_2H_2 + O_2 = 2CO + H_2 + \text{HEAT}$$ (First stage; inner core)

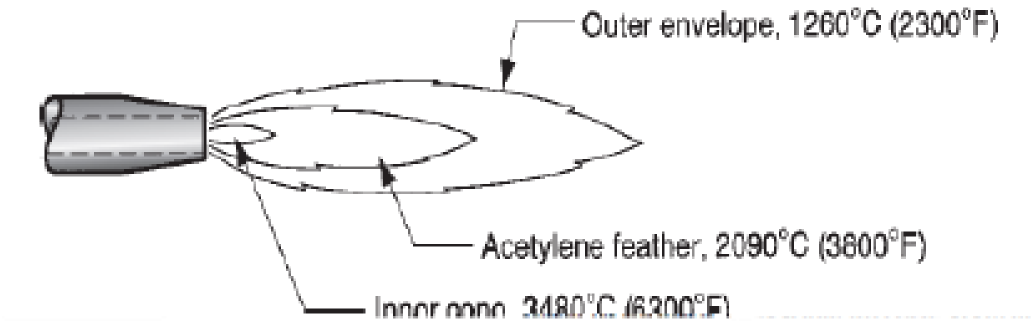
The products of first reaction are combustible and second reaction occurs as,

$$2CO + H_2 + 1.5O_2 = 2CO_2 + H_2O + \text{Heat}$$ (second stage; outer envelope)

When both oxygen and acetylene are mixed in ratio of 1:1, then neutral flame is seen as shown in figure. The outer envelope delivers a temperature of 1260°C and inner core has app. 3500°C.

The first stage reaction is seen as the inner cone of the flame (bright white colour), while the second stage reaction is seen in the outer envelope (colorless but with tinges ranging from blue to orange). The temperature is very high at the inner core which is app. 3500°C.

Total heat liberated during the two stages of combustion is 55×10^6 J/m^3 of acetylene. But the heat transfer factor in OAW is 0.1 to 0.3 as the flame spreads over large region.



Neutral flame in oxyacetylene welding

Problems and advantages of OAW:

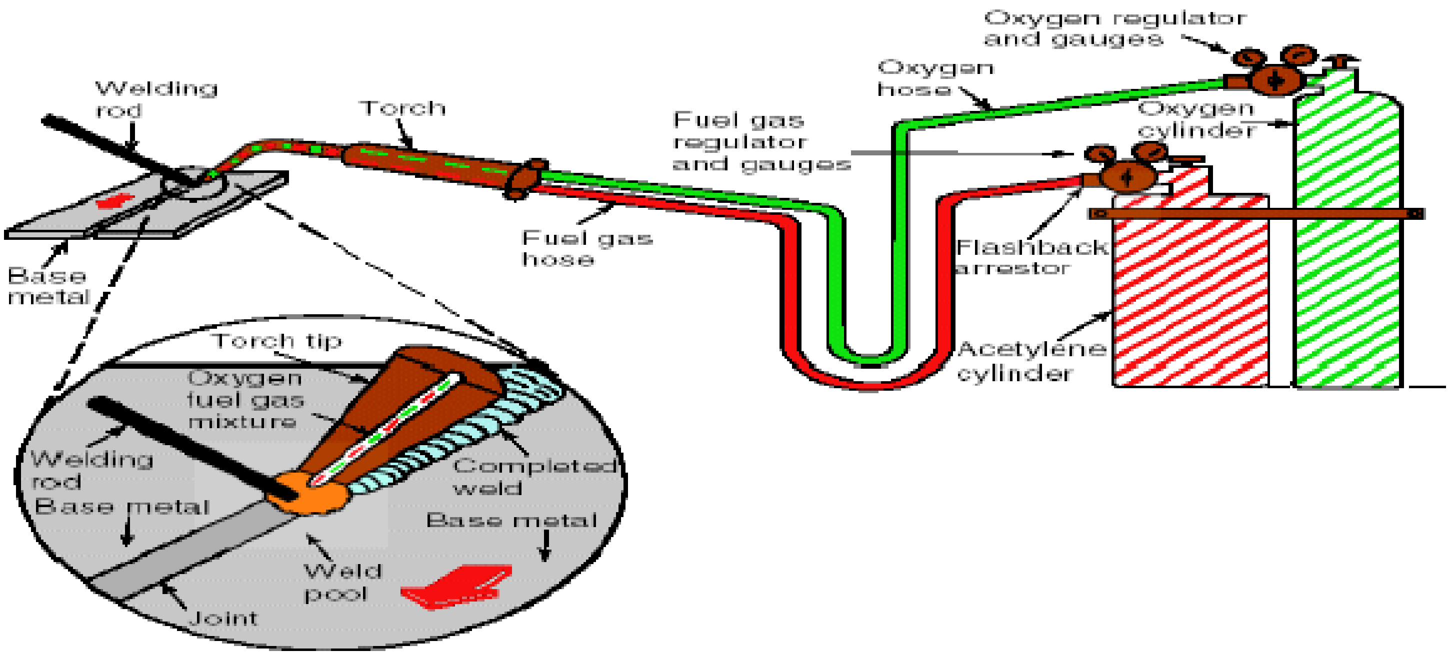
• The combination of acetylene and oxygen is highly flammable and hence hazardous to environment.

• It is unstable at pressures much above 1 atm

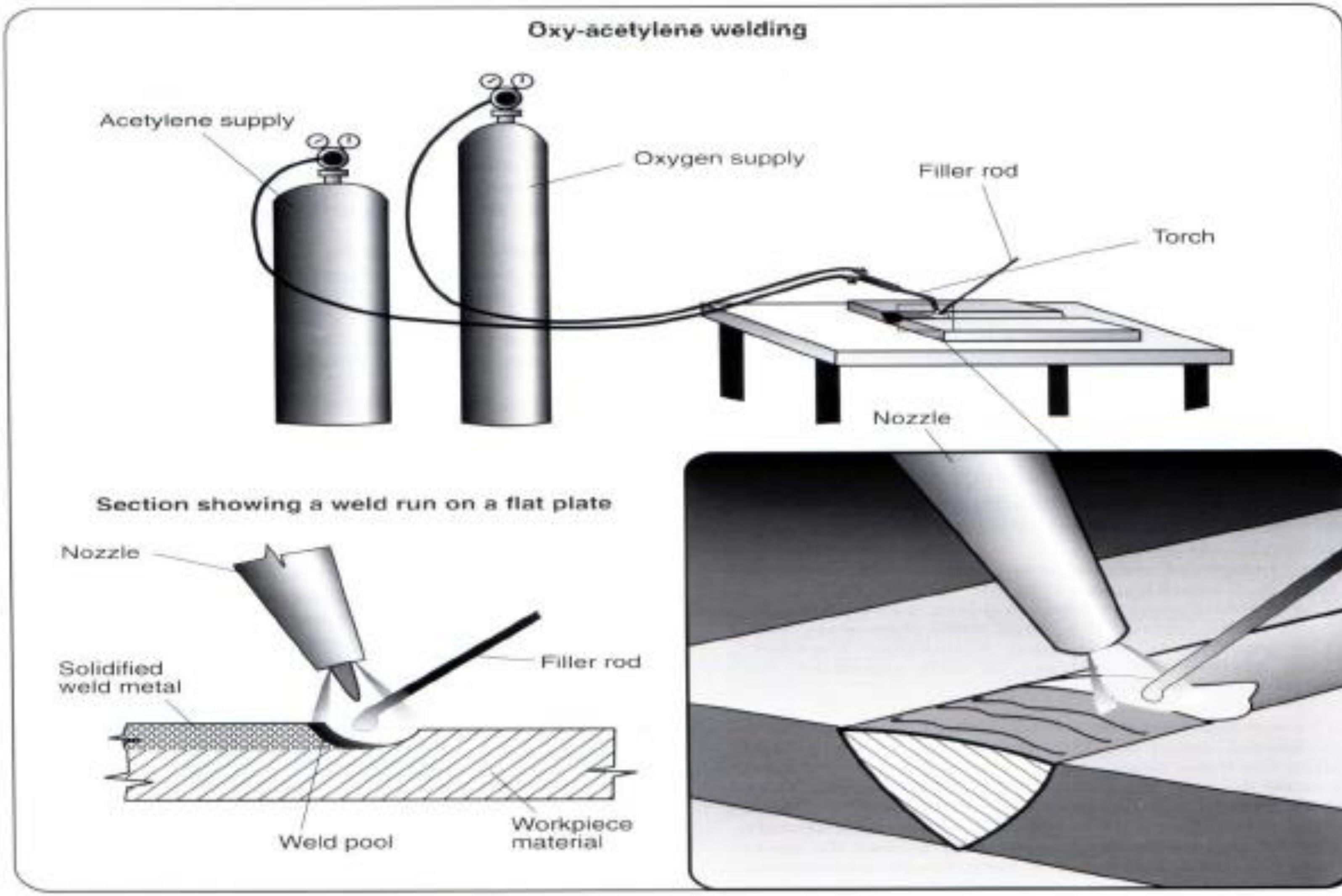
• It is mandatory for the welder to wear gloves, goggles etc. as preventive measures.

• The equipment is relatively cheap and portable. So it is used as an economical, versatile process that is well suited for low quantity production and repair jobs.

• It is rarely used to weld plates thicker than 6.5 mm.

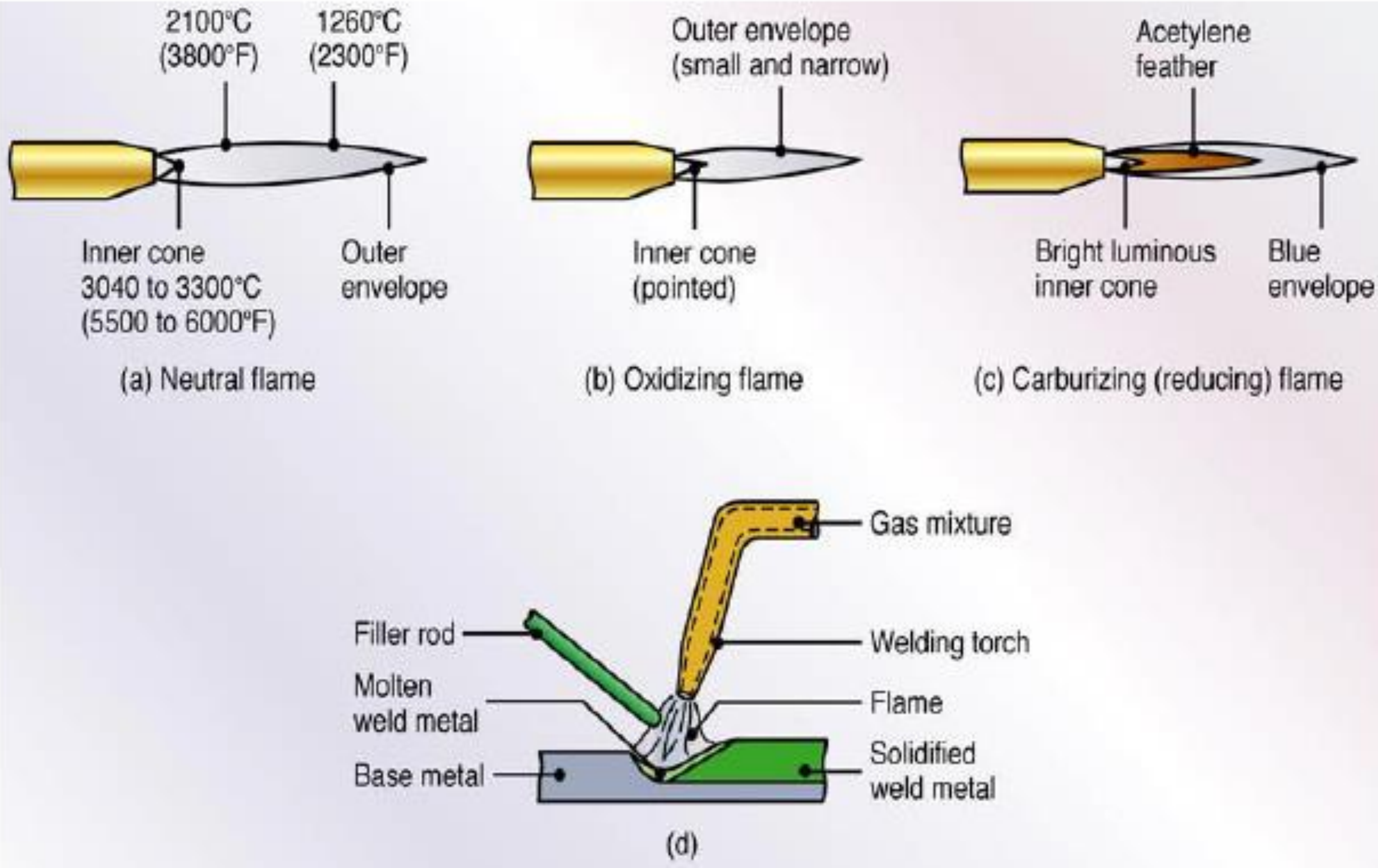


Gas Welding



Oxy-acetylene welding
Acetylene supply
Oxygen supply
Filler rod
Torch
Nozzle
Section showing a weld run on a flat plate
Nozzle
Filler rod
Solidified
weld metal
Weld pool
Workpiece
material

Oxy-acetylene flames



(a) Neutral flame (b) Oxidizing flame (c) Carburizing (reducing) flame

(d)

The Oxy-acetylene welding Flame

Carburising

Neutral

Oxidising

Types of flames use in gas welding

- There are three distinct types of oxy-acetylene flames, usually termed:
  - Neutral
  - Carburizing (or “excess acetylene”)
  - Oxidizing (or “excess oxygen”)
- The type of flame produced depends upon the ratio of oxygen to acetylene in the gas mixture which leaves the torch tip.

Types of flames

1)Neutral flame

It is produced when the ratio of oxygen to acetylene, in the mixture leaving the torch, is almost exactly one-to-one. It's termed "neutral" because it will usually have no chemical effect on the metal being welded. It will not oxidize the weld metal; it will not cause an increase in the carbon content of the weld metal.

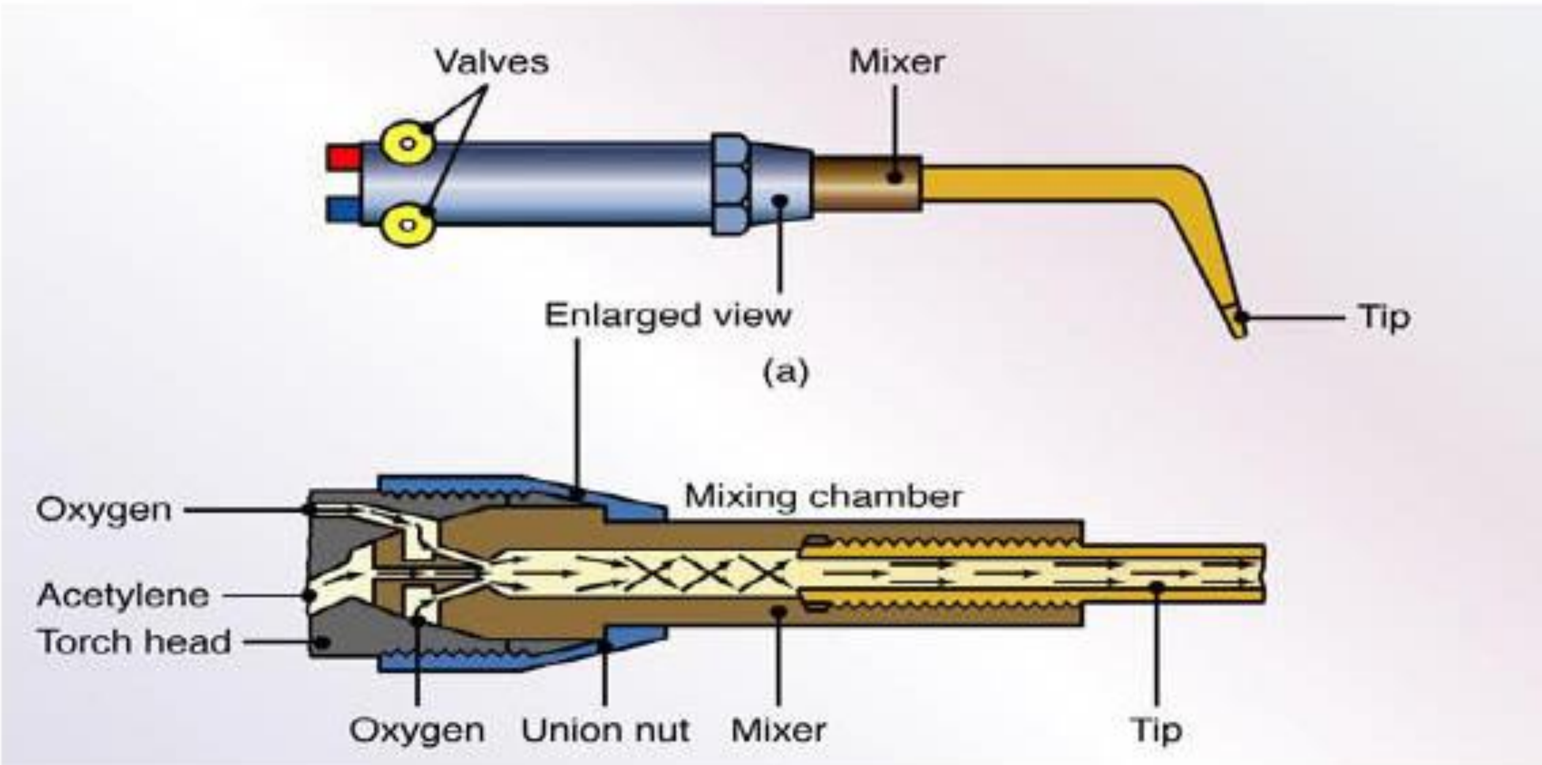
2)excess acetylene flame/ Carburising flame

as its name implies, is created when the proportion of acetylene in the mixture is higher than that required to produce the neutral flame. Used on steel, it will cause an increase in the carbon content of the weld metal.

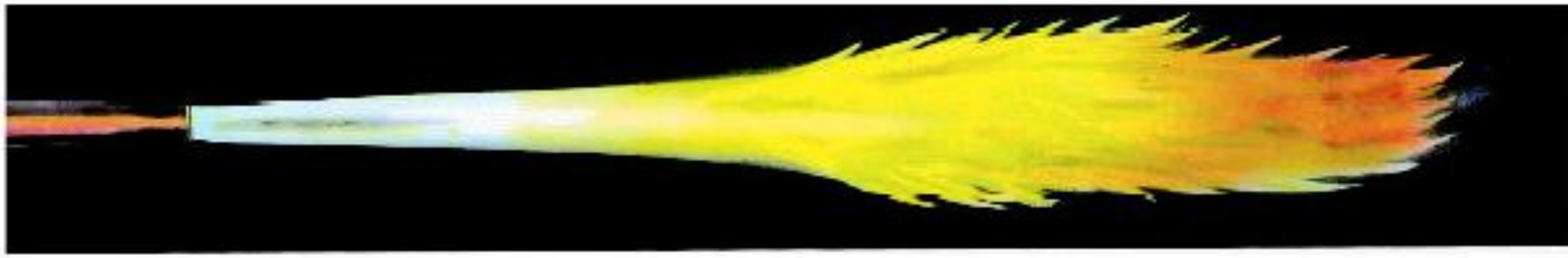
3)Oxidizing flame

results from burning a mixture which contains more oxygen than required for a neutral flame. It will oxidize or "burn" some of the metal being welded.

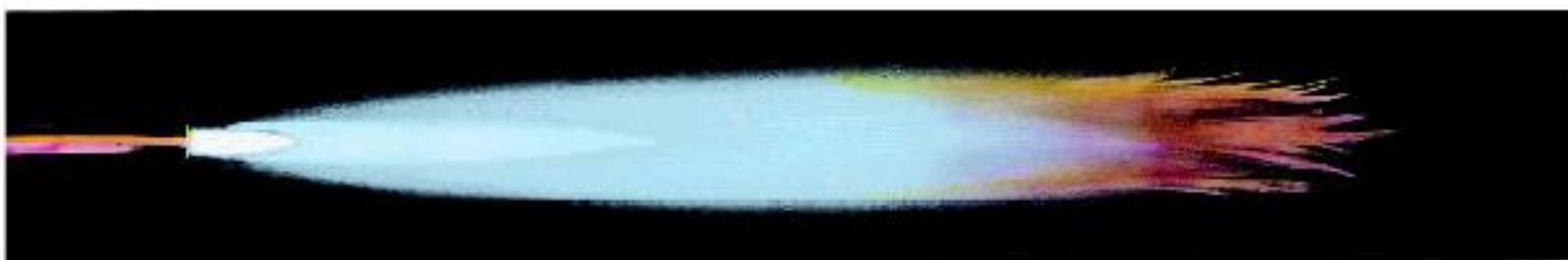
Typical torch styles:-



Pure Acetylene and Carburizing Flame profiles



Acetylene Burning in Atmosphere
Open fuel gas valve until smoke clears from flame.



Carburizing Flame
(Excess acetylene with oxygen.) Used for hard-facing and welding white metal.

Neutral and Oxidizing Flame Profiles

Neutral Flame
(Acetylene and oxygen.) Temperature 5589°F (3087°C). For fusion welding of steel and cast iron.

Oxidizing Flame
(Acetylene and excess oxygen.) For braze welding with bronze rod.

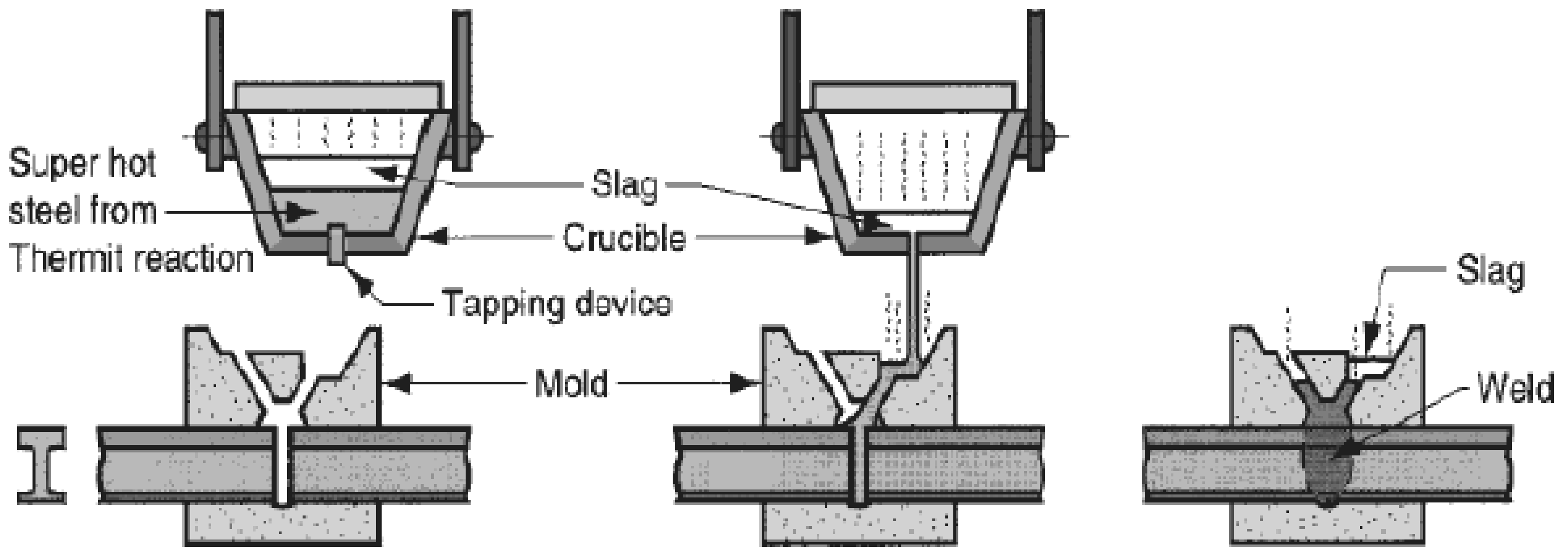
Thermit welding

Thermite (thermit): a mixture of aluminum powder and iron oxide that produces an exothermic reaction when ignited.

In thermit welding, the heat for coalescence/joining is produced by superheated molten metal formed from the chemical reaction of thermit.

The following chemical reaction is seen when a thermit mixture is ignited at 1300°C. The temperature of the reaction is 2500°C.

$$8Al + 3Fe_3O_4 = 9Fe + 4Al_2O_3 + heat$$



- At this temperature, superheated molten iron plus aluminum oxide is made that floats on the top as a slag and protects the iron from the atmosphere.

- **Applications of TW:**
- Joining of railway lines, repair of cracks in large steel castings and forgings like ingot molds, large diameter shafts, frames for machinery etc.
- This process can be used to weld heavy parts on site.
- It is useful for welding heavy sections.

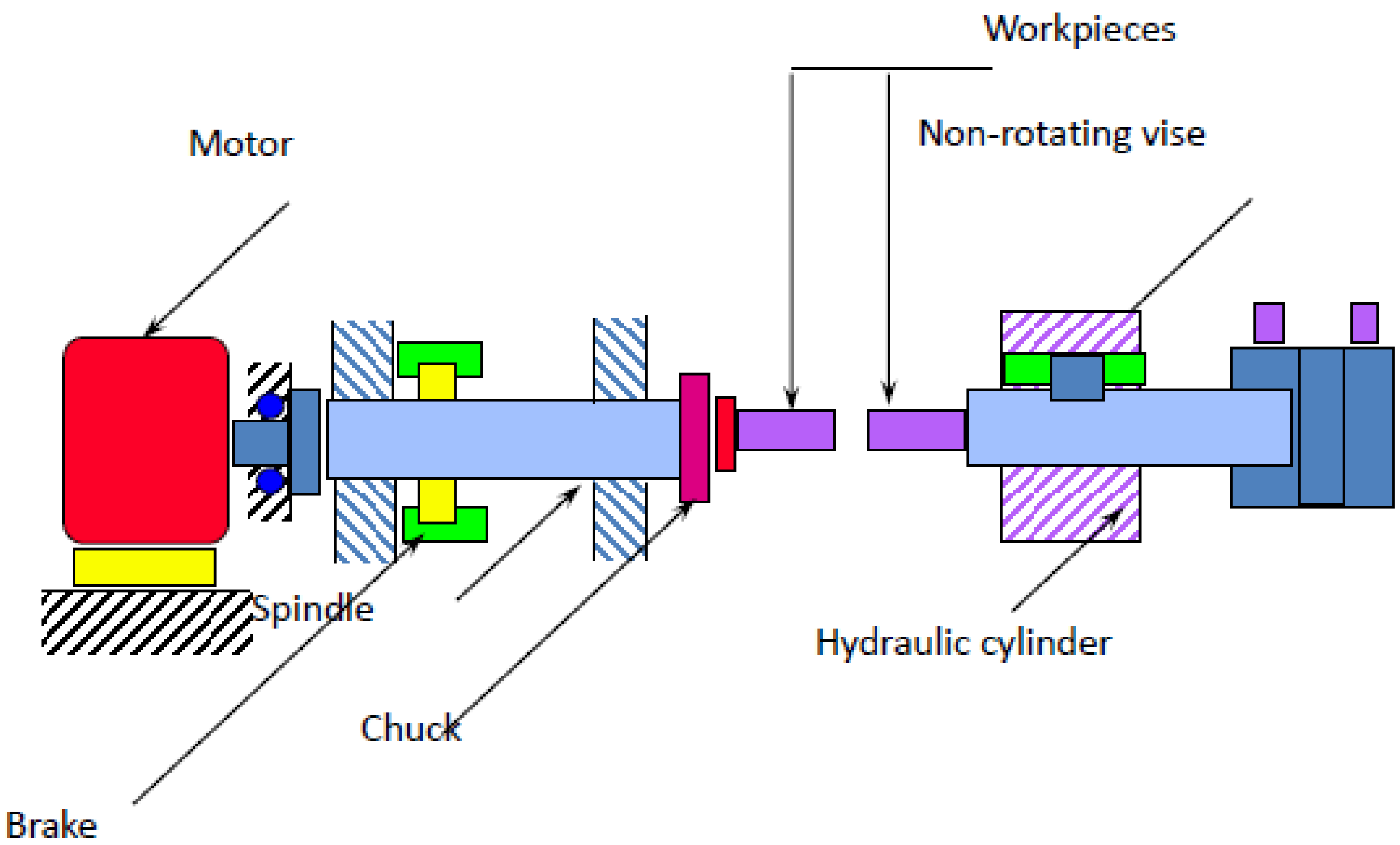
Friction welding

-Friction heat caused by the motion of one surface against another enables plastic deformation and atomic diffusion at the interface

-Used by the automotive industry for decades in the manufacture of a range of components

-The weld is formed across the entire cross-sectional area of the interface in a single shot process

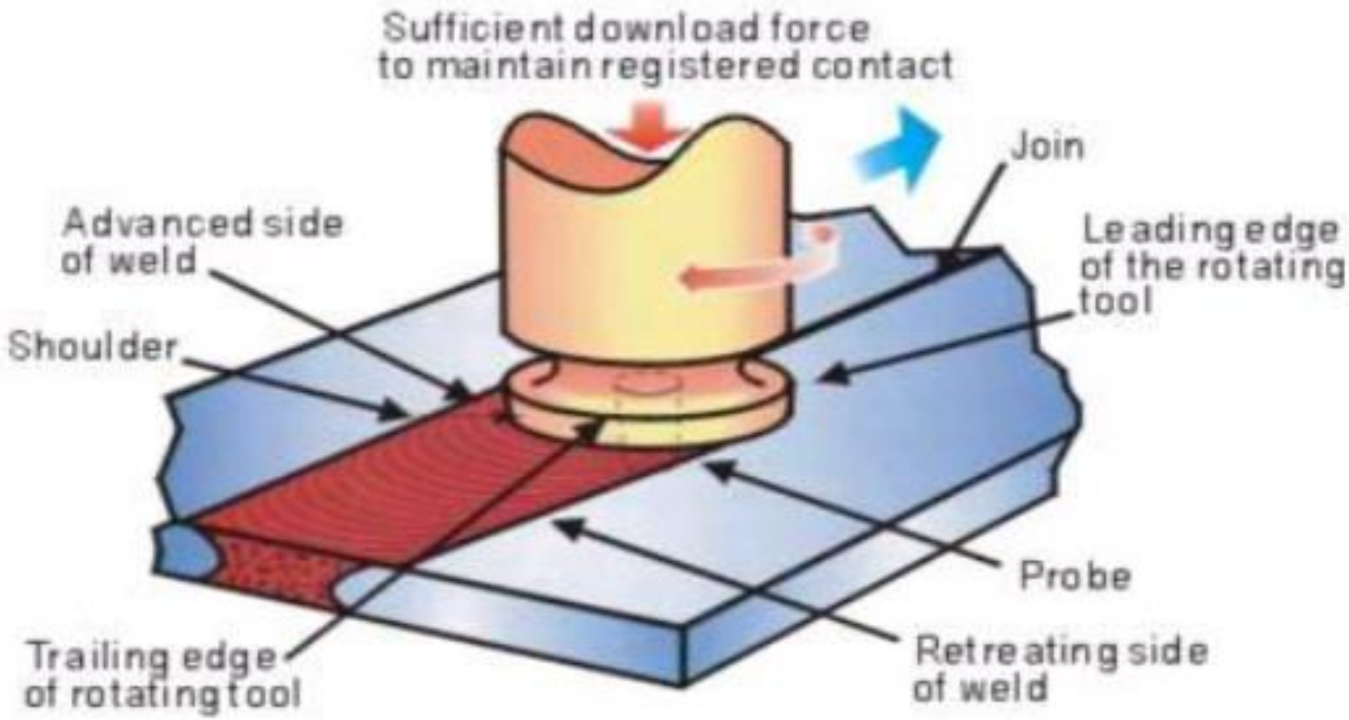
Friction welding



Advantages of friction welding

- Narrow HAZ
- Dissimilar metals can be joined
- No fusion zone
- Can be used under water
- very high reproducibility - an essential requirement for a mass production industry
- Excellent weld quality, with none of the porosity that can arise in fusion welding
- environmentally friendly, because no fumes or spatter are generated, and there is no arc glare or reflected laser beams with which to contend

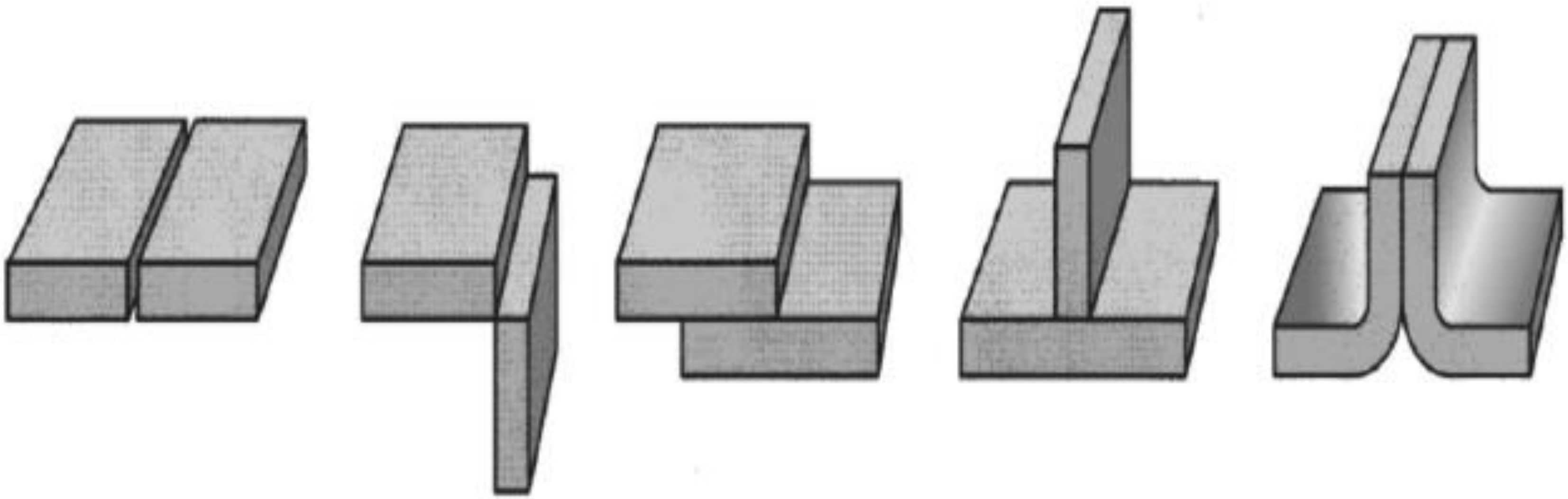
Friction surface / Friction stir welding



Steps in friction stir welding

- A non-consumable rotating tool is pushed into the materials to be welded and then the central pin, or probe, followed by the shoulder, is brought into contact with the two parts to be joined.
- The rotation of the tool heats up and plasticises the materials it is in contact with and, as the tool moves along the joint line, material from the front of the tool is swept around this plasticised annulus to the rear, so eliminating the interface.

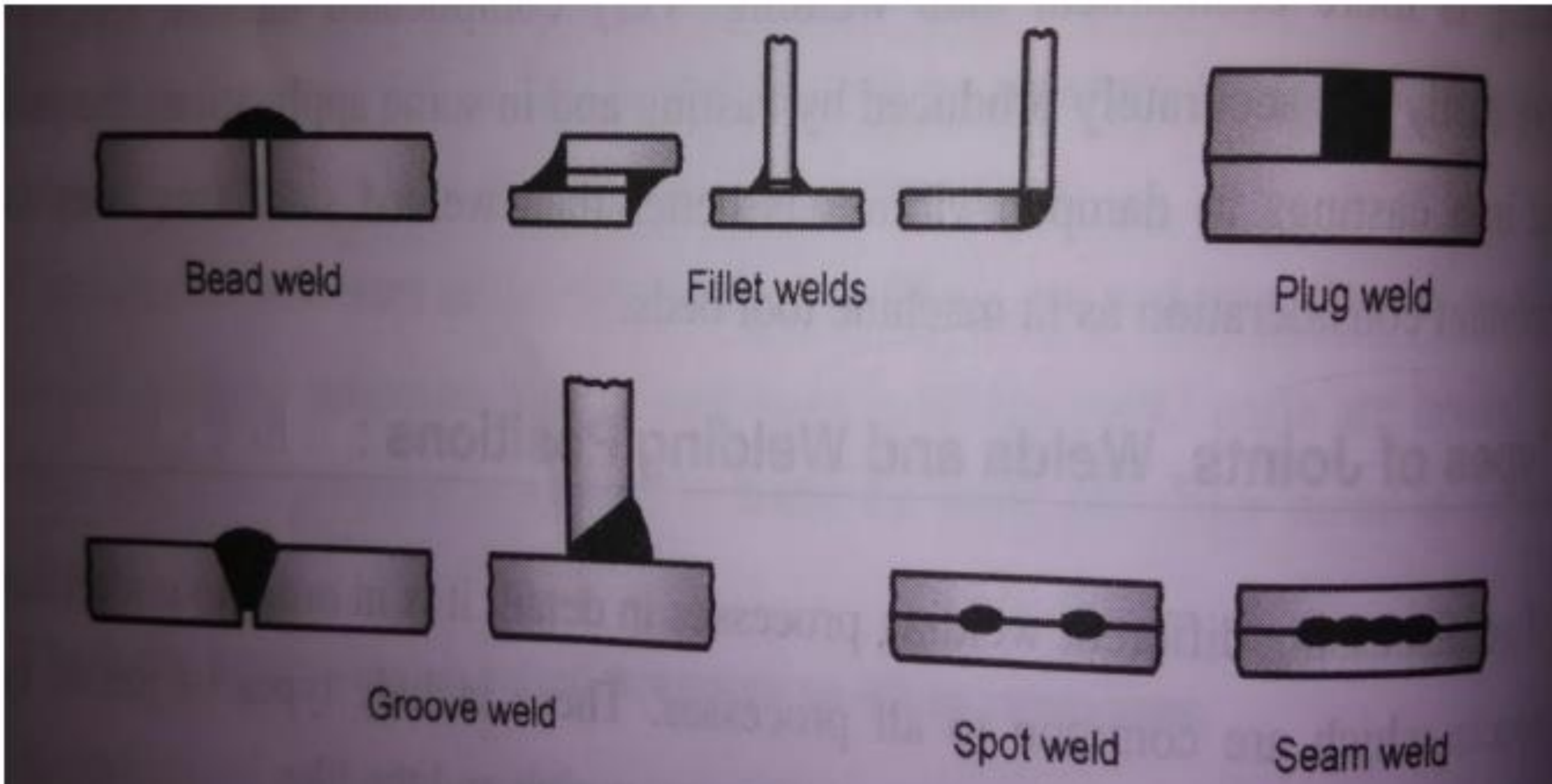
Types of Weld joint

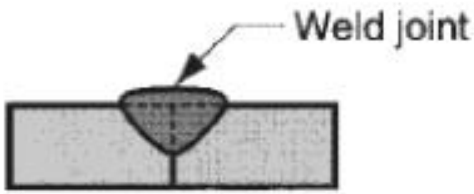


Butt joint Corner joint Lap joint Tee joint Edge joint

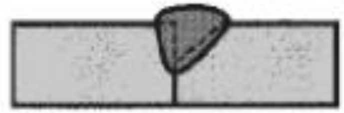
Types of Welds

- The type of weld depends on the joint.
- As shown in fig. weld may be a bead weld, fillet weld, plug weld, groove weld, spot weld or seam weld

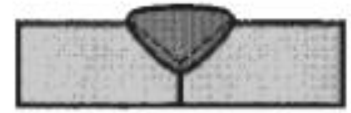




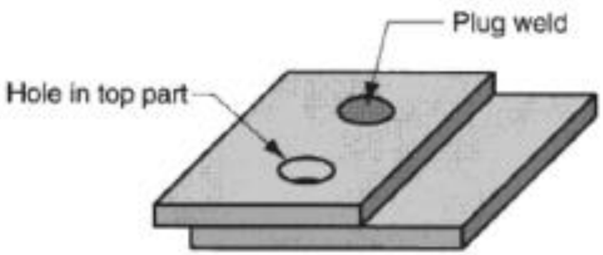
square groove weld one side



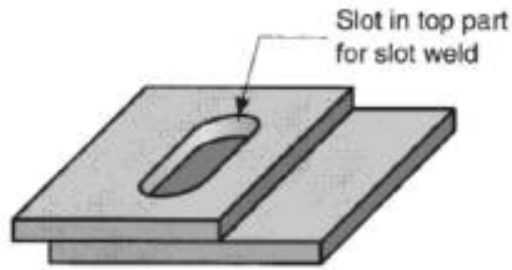
single bevel groove weld



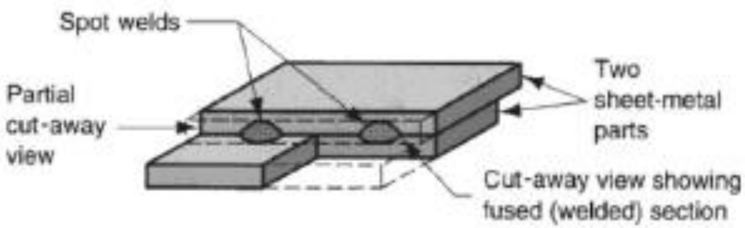
single V-groove weld



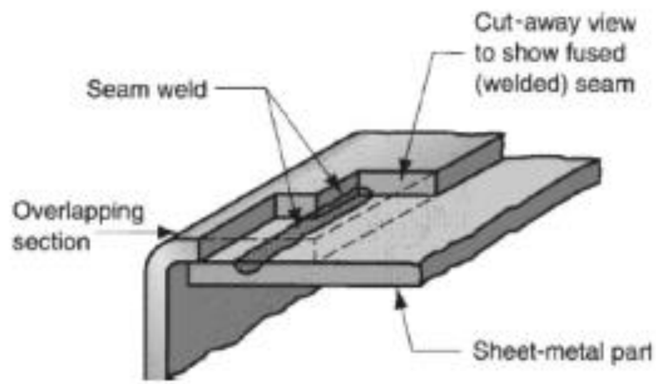
Plug weld



Slot weld



Spot weld



Seam weld

Welding defects:-

Defects in Weldments (unit formed by welding together an assembly of pieces)

- Defects occur in weldments due to improper welding procedures or due to random causes. With proper care these defects can be prevented.
- The defects commonly occurring can be classified into 3 main categories:

  1. Dimensional defects.

  2. Structural discontinuities.

  3. Inadequate properties.

1. **Dimensional defects:** Warpage, incorrect joint preparation, incorrect weld size and incorrect profile of the weld.
2. **Structural discontinuities:** Porosities, inclusion, incomplete fusion, under welding, inadequate penetration, cracks and other surface defects.
3. **Inadequate properties:** Low tensile strength, low yield strength, low ductility, inadequate hardness, impact failure, incorrect composition, and improper corrosion resistance.

- These effects may be further classified into **external** and **internal** defects.

- Defects such as incorrect profile, crater, melted edge, surface porosity are visible on the surface and are called **surface** or **external defects.**
- **Internal defects** or **cracks** are not visible on the surface.
- These include blow holes, deep cracks, inclusions and incomplete penetration.
- The following defects are commonly found in weldments:

1. Undercut.
2. Incomplete fusion.
3. Porosity.
4. Slag inclusion.
5. Weld cracking.
6. Voids and Craters
7. Distortion.
8. Corrosion.

- **Undercut:** Undercut is a small notch at the weld interface. It is caused by too high welding current and improper welding technique. It is more likely in horizontal and vertical welding.
- **Incomplete Fusion:** This is caused by insufficient penetration of the joint, incorrect welding technique, wrong design of the joint, or poor selection of welding parameters and improper cleaning of the joint.

- **Porosity:** Porosity is caused by entrapment of gases during the solidification process. The gases mostly consists of hydrogen, oxygen, nitrogen of which hydrogen is most prominent for causing porosity. Gases like argon, helium or carbon dioxide do not cause porosity because they are insoluble.

- **Slag Inclusion:** Slag is formed by the reaction of fluxes and is expected to float out at the top of molten metal and be removed after solidification. Slag may also be present in multi-pass welds.

- **Weld cracking:** Weld cracking may be hot cracking or cold cracking. Hot cracking occurs during the root pass if the mass of the base metal is very large compared to the weld metal deposited. It can be controlled by preheating the base metal, by changing the contour, composition of weld bead.

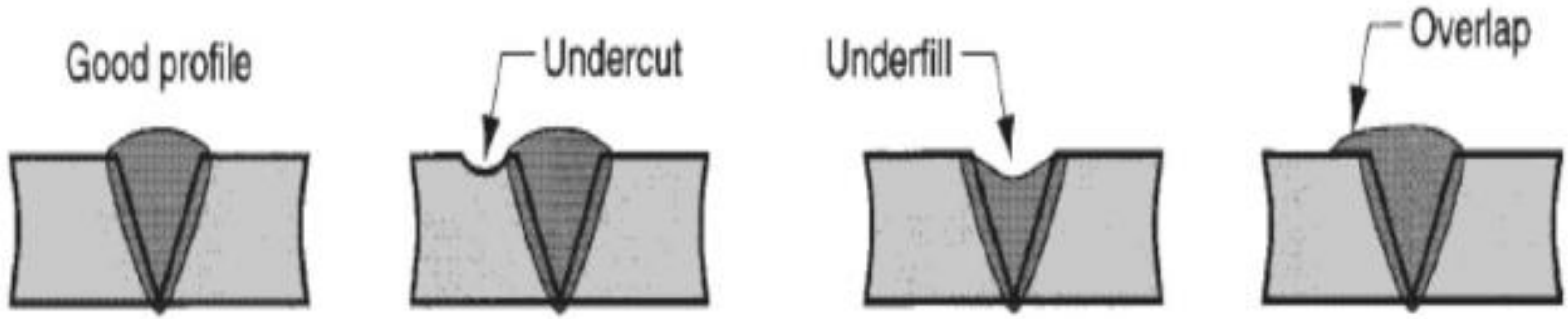
- **Voids and Craters:** It has been shown that voids upto 7% of the cross section have not much effect on the tensile or impact strength or the ductility of the weld. If the size of voids are larger presence of foreign matters cause a large reduction in the strength of the weld leading to opening of cracks.

- **Distortion:** Distortion is one of the major problems found in weldments. It is caused mainly by shrinkage. 3 types of distortion are possible in weldments:

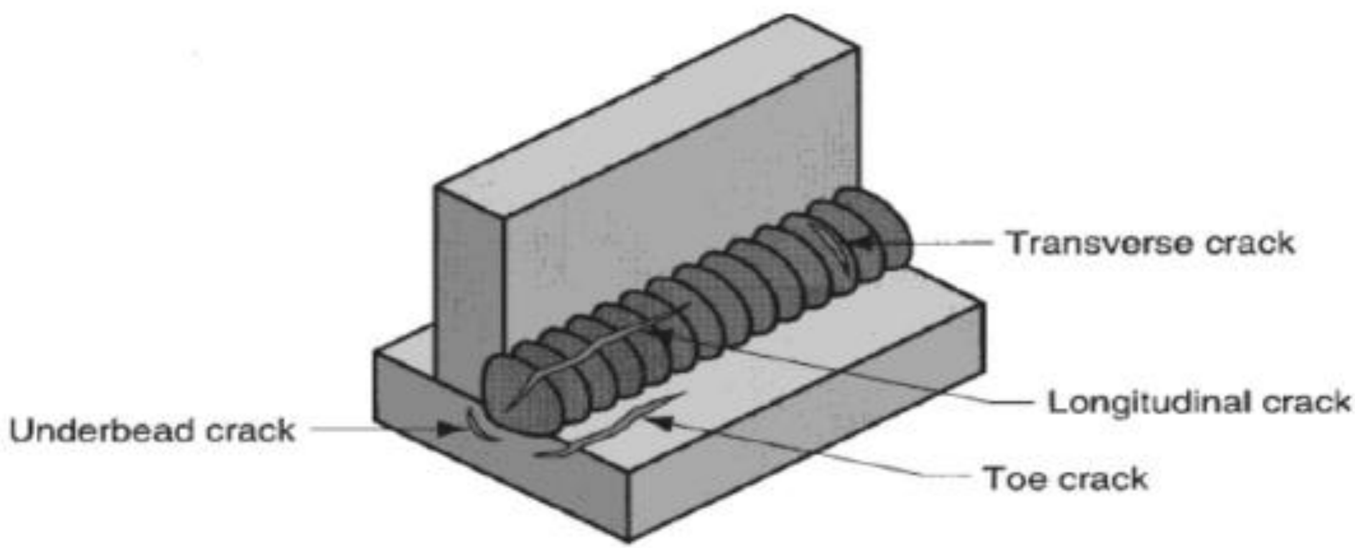
1. Longitudinal Shrinkage: This occurs parallel to the weld line and is so small that it can be ignored.
2. Transverse shrinkage: This occurs perpendicular to the weld line. It is result of contraction of base metal which had expanded during welding.
3. Angular change or orientation about the weld line: The weld distortions occur because of the shrinkage that takes place in weldments. It cannot be completely eliminated but can be reduced by restraining the pieces being joined so that the distortions cannot take place.

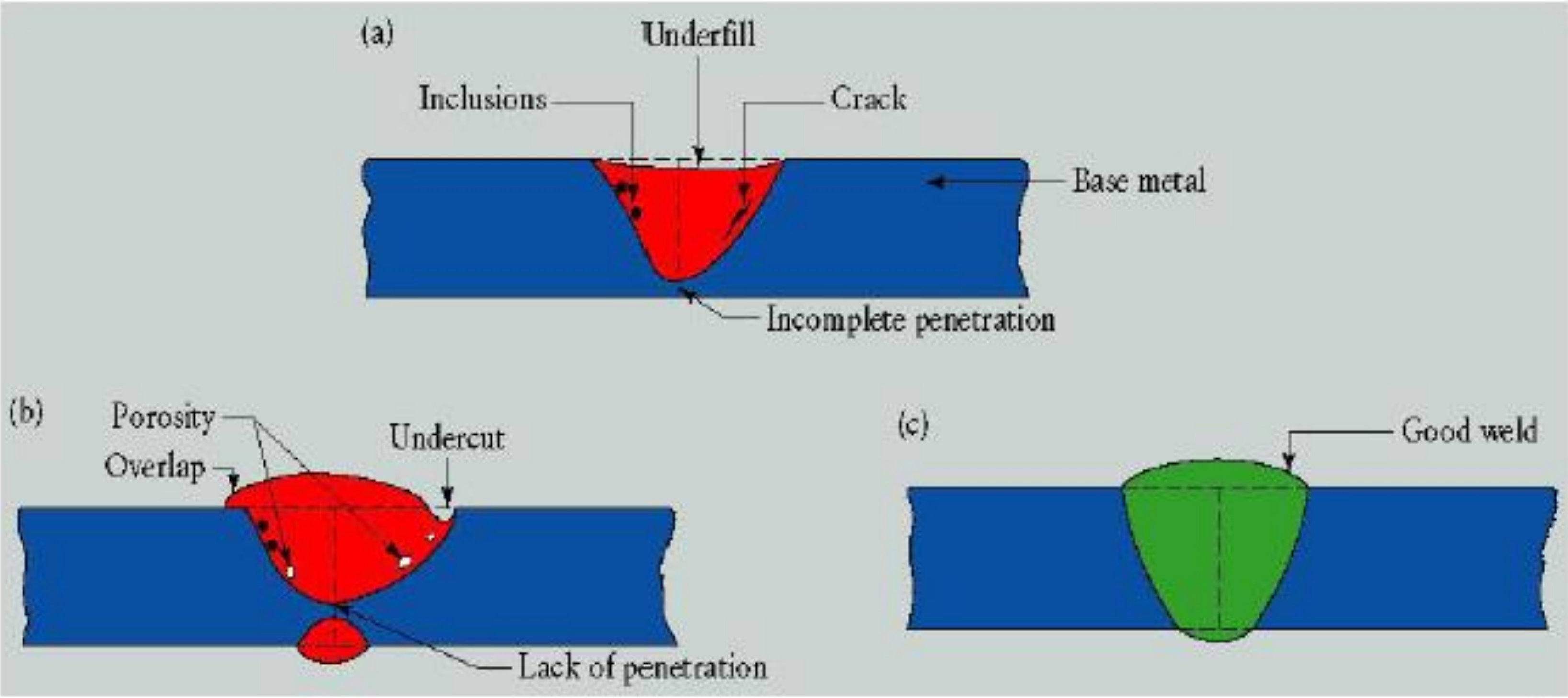
- **Corrosion:** Welding makes metals more susceptible to corrosion in a number of ways. The intense heat of welding removes protective coatings from metal surfaces and also changes some metals to make them more susceptible to corrosion. For ex: welding can make stainless steel lose its corrosion resistance.

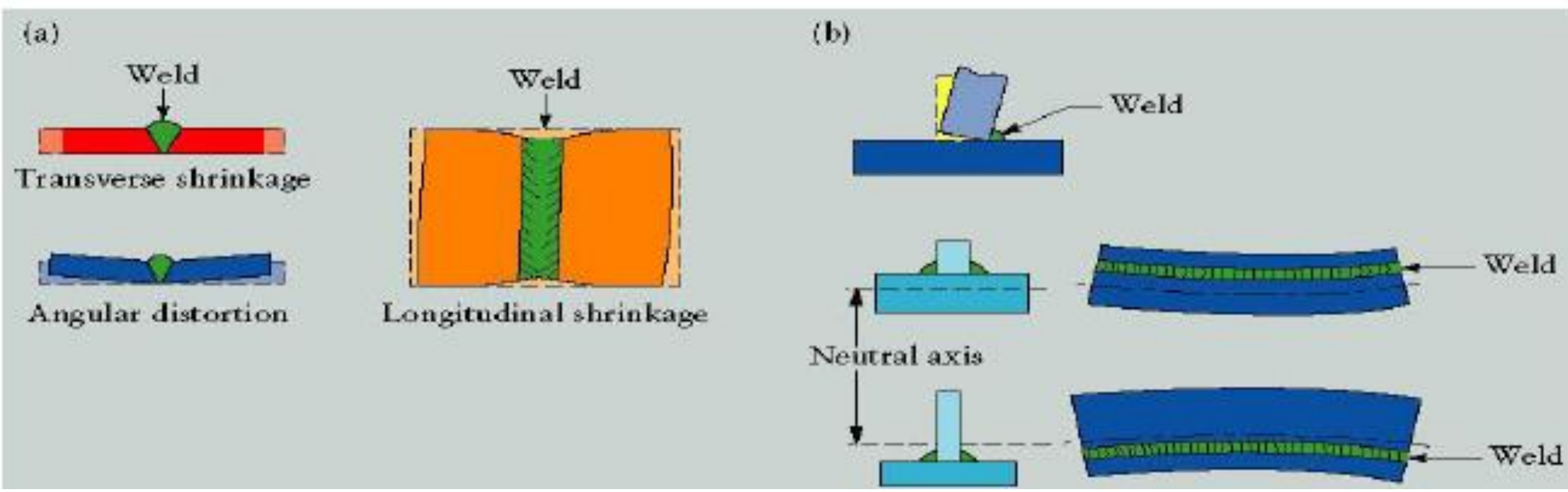
Improper weld profile:



Weld cracks:

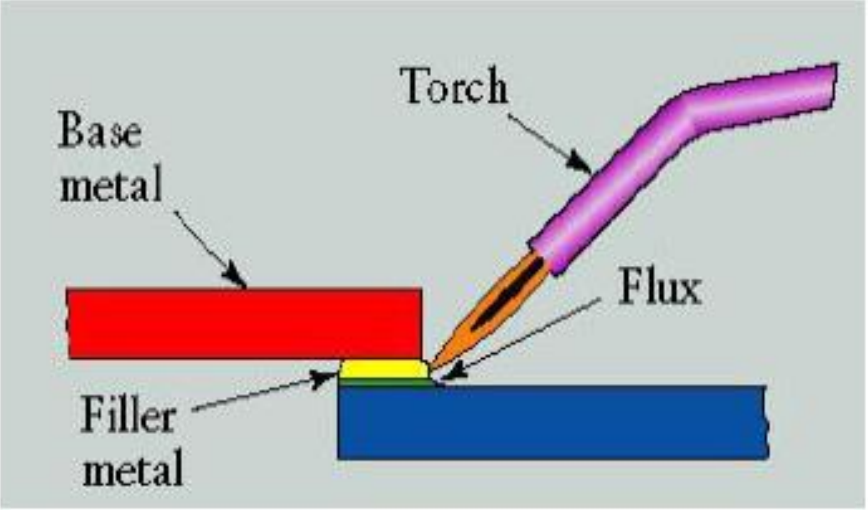



(a)
Underfill
Inclusions
Crack
Base metal
Incomplete penetration
(b)
Porosity
Undercut
Overlap
Lack of penetration
(c)
Good weld


(a)
Weld
Transverse shrinkage
Angular distortion
Weld
Longitudinal shrinkage
(b)
Weld
Weld
Neutral axis
Weld

Brazing

- It is a joining process in which a filler metal is melted and distributed by capillary action between the faying (contact) surfaces of the metal parts being joined.
- In brazing, the filler metal has a melting temperature above 450°C, but below the melting point of base metals to be joined.
- Join produced by this welding is stronger than soldering.
- This process offers better corrosion resistance.
- Filler used in brazing include Cu and Cu alloys, silver alloys and Al alloys.
- In this process heating is done by torch, furnace, induction, resistance, bath dipping infrared techniques.



Advantages of brazing

- Brazing can be used to join a large variety of dissimilar metals.
- Pieces of different thickness can be easily joined by brazing
- Thin-walled tubes & light gauge sheet metal assemblies not joinable by welding can be joined by brazing.
- Complex & multi-component assemblies can be economically fabricated with the help of brazing.
- Inaccessible joint areas which could not be welded by gas metal or gas tungsten arc spot or seam welding can be formed by brazing.

- **Applications:**

1)Automobile – Joining Tubes

2) Pipe/Tubing joining (HVAC)

3) Electrical equipment - joining wires

4) Jewelry Making

SOLDERING

- Soldering is similar to brazing and can be defined as a joining process in which a filler metal with melting point not exceeding 450°C is melted and distributed by capillary action between the faying surfaces of the metal parts being joined.
- As in brazing, no melting of the base metals occurs, but the filler metal wets and combines with the base metal to form a metallurgical bond.
- Filler metal, called Solder, is added to the joint, which distributes itself between the closely fitting parts.
- Strength of the joint is weak
- Corrosion resistance is less

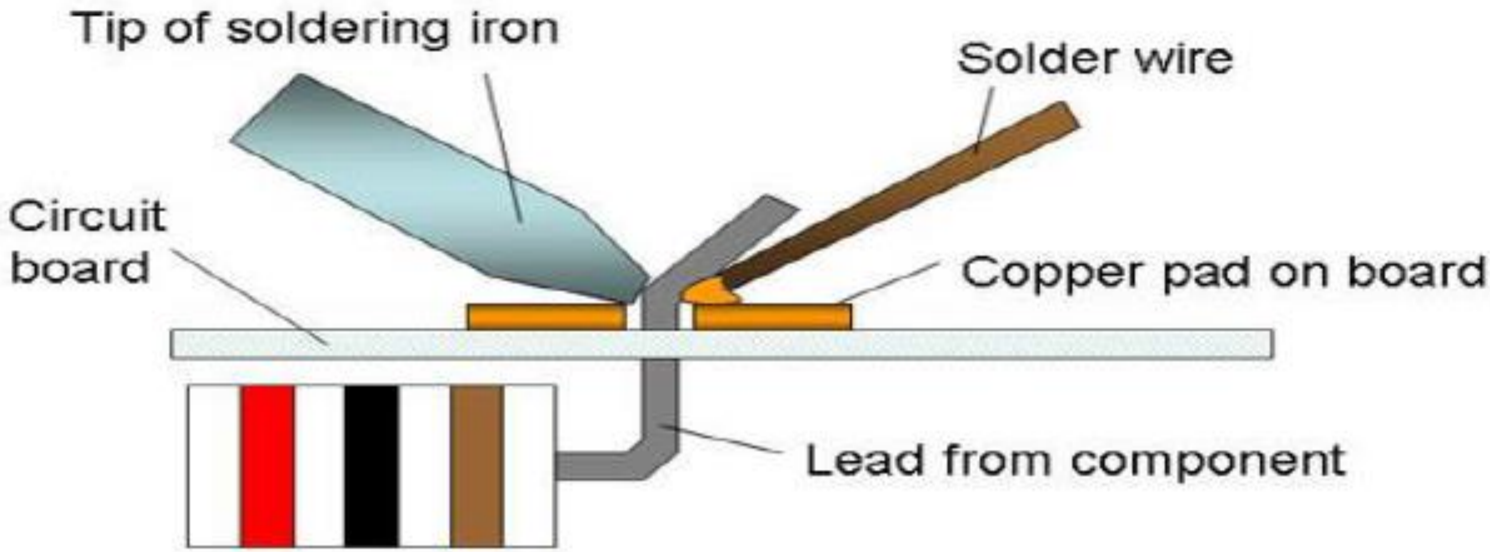
SOLDER: Alloys of Tin and Lead. Tin is chemically active at soldering temperatures and promotes the wetting action required for successful joining.

Applications:

1) Printed Circuit Board (PCB) manufacture
2) Pipe joining (copper pipe)

Easy to solder: copper, silver, gold

Difficult to solder: aluminum, stainless steels


Tip of soldering iron
Solder wire
Circuit board
Copper pad on board
Lead from component

Comparison between Welding,soldering and brazing

Sl. No.	Welding	Soldering	Brazing
1.	These are the strongest joints used to bear the load. Strength of a welded joint may be more than the strength of base metal.	These are weakest joint out of three. Not meant to bear the load. Use to make electrical contacts generally.	These are stronger than soldering but weaker than welding. These can be used to bear the load upto some extent.
2.	Temperature required is upto 3800°C of welding zone.	Temperature requirement is upto 450°C.	It may go to 600°C in brazing.
3.	Workpiece to be joined need to be heated till their melting point.	No need to heat the workpieces.	Workpieces are heated but below their melting point.

4.	Mechanical properties of base metal may change at the joint due to heating and cooling.	No change in mechanical properties after joining.	May change in mechanical properties of joint but it is almost negligible.
5.	Heat cost is involved and high skill level is required.	Cost involved and skill requirements are very low.	Cost involved and sill required are in between others two.
6.	Heat treatment is generally required to eliminate undesirable effects of welding.	No heat treatment is required.	No heat treatment is required after brazing.
7.	No preheating of workpiece is required before welding as it is carried out at high temperature.	Preheating of workpieces before soldering is good for making good quality joint.	Preheating is desirable to make strong joint as brazing is carried out at relatively low temperature.